How Does Trade Competition with a Technological Leader Reshape Domestic Innovation? Evidence from South Korea

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Abstract

Firms facing import competition may change not only how much they innovate but also *where*. Korea's abolition of Japan-specific import restrictions on selected products exposed domestic firms to competition from a technological leader, raising imports of these products from Japan by 66 percent. At the technology level, patenting in exposed fields fell by 19 percent, with larger declines where Japan was initially further ahead. Yet firm-level evidence reveals substantial reallocation across technologies, with little change in incumbents' total patenting. Firms with more prior experience outside the affected fields reallocated more strongly and sustained total patenting; those concentrated in and around the exposed technologies reallocated less and gradually reduced total patenting. The reallocation favored technologies that had been gaining momentum abroad before liberalization.

Keywords: Trade Liberalization, Import Competition, Innovation, Industrial Policy, Technology Gap

JEL Classification: F13, F14, O25, O31, O33

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1 Introduction

Successful late industrialization has usually begun under protection. Its withdrawal can expose domestic firms to competition from the technological leaders they had been trying to catch. Do firms respond by intensifying innovation in the exposed technology or by retreating from it? If they retreat, does this reflect an erosion of innovative capacity or a reallocation toward other fields? Firms rarely innovate in one technology alone. A retreat confined to the protected technology and a broad decline in innovative activity carry different implications for the withdrawal of protection, but technology-level evidence alone cannot distinguish them.

The answer may turn on technological distance and the location of prior capabilities. A larger gap from the foreign leader makes successful catch-up less likely and lowers the expected return to research in the exposed technology. Prior experience elsewhere then shapes the response: firms can redirect research toward fields in which they have established capabilities, while those concentrated in and around the exposed technologies have less scope to reallocate. The same shock can therefore leave one firm's total innovative activity largely intact while reducing another's.

South Korea's Import Source Diversification Program (ISDP) offers a sharp setting in which to study these responses. Beginning in 1978, the program barred designated products from being imported when their country of origin was Japan, while leaving imports from other sources unrestricted. Following the Asian Financial Crisis, Korea committed under its IMF program to abolishing the remaining restrictions, and the final 43 products were liberalized in 1998 and 1999. The abolition raised imports of these products from Japan by roughly 66 percent. It generated a source-specific competitive shock from a technological leader whose initial advantage over Korea varied considerably across the affected technologies.

I combine bilateral trade data, Korean patent applications, and establishment-level manufacturing data to trace adjustment from import competition to production and innovation. I link the final 43 products to associated technologies and compare exposed and unexposed products and technologies within narrow categories before and after abolition, allowing their pre-liberalization characteristics to predict different time paths. This comparison accounts for differences in initial competitiveness and production structure as well as shocks shared by related products and technologies. To measure Japan's initial quality advantage, I combine Japanese and Korean sales and unit values for the same detailed products in the United States in 1996–1997, following the demand-based approach of Khandelwal et al. (2013).

I document three main findings. First, Korean patenting in the exposed technologies fell by approximately 19 percent following liberalization. Domestic production and employment of the affected final goods also declined, but more gradually, with larger production losses appearing in the mid-2000s. Patenting thus adjusted before the full production response materialized, suggesting that firms reduced innovation in anticipation of lower returns.

Second, the decline in patenting was concentrated in technologies where Japan held a larger initial quality lead. Within product-linked patent groups, filings fell by approximately 26 percent when the initial quality gap was above the median, compared with 14 percent below it. This quality gradient is sharper and statistically significant when products are linked directly to granular CPC technologies. Measuring quality relative to Japan connects the innovation response to domestic products' position against the competitor gaining market access.

Third, how firms adjust depends on the location of their prior knowledge. Incumbents with substantial patenting experience outside the affected fields show no contraction in overall patenting while reallocating research toward other fields. Firms whose knowledge is concentrated in and around the exposed technologies reallocate less and show a gradual contraction in total patenting. Portfolio-share regressions further link the destinations of reallocation to pre-liberalization patent growth abroad, suggesting adjustment toward expanding international technological opportunities. A decline in patents associated with protected technologies can therefore reflect a change in firms' research direction, so technology-specific losses alone provide an incomplete measure of the innovation consequences of withdrawing protection.

The findings contribute to three related literatures. First, they add to research on competition, innovation, and technological distance. Competition can reduce post-innovation rents while increasing the incentive to escape close rivals (Aghion et al., 2005; Akcigit et al., 2026), and empirical studies find both positive and negative innovation responses to import competition (Bloom et al., 2016; Autor et al., 2020; Liu et al., 2021; Ahsan and Chakraborty, 2024). Much of this evidence concerns competition from lower-cost exporters. I study a catching-up economy exposed to a technological leader. Prior work shows that foreign entry and trade liberalization have different effects depending on distance from the technological frontier (Aghion et al., 2009; Amiti and Khandelwal, 2013). The bilateral quality measure connects that distance to the specific foreign competitor responsible for the shock.

Second, the paper contributes to research on the direction of innovation and firms' technological portfolios. Economic incentives redirect research across product characteristics, production technologies, and clean

and dirty innovation (Newell et al., 1999; Hanlon, 2015; Aghion et al., 2016; Noailly and Smeets, 2015; Branstetter et al., 2026). Such movement depends on the knowledge firms already possess (Jaffe, 1986; Noailly and Smeets, 2015). I show how prior experience outside the affected fields shapes adjustment to import competition: firms with an outside-field knowledge base reallocate more strongly while maintaining total patenting. This response parallels the product-mix adjustment of multi-product firms after trade liberalization (Bernard et al., 2010, 2011; Mayer et al., 2014) and complements evidence on the reallocation of aggregate research across firms (Gong and Xu, 2017).

Finally, the paper contributes to research on industrial policy and economic development by examining the withdrawal of protection, an aspect of policy design emphasized by Juhász et al. (2024). Temporary protection can generate persistent industrial capabilities through learning and spillovers (Juhász, 2018). Research on South Korea emphasizes support for foreign technology adoption, the transition to domestic innovation, and the role of industrial policy and large productive firms in industrialization (Choi and Shim, 2026, 2024; Lane, 2025; Choi et al., 2024). I examine how innovative activity in previously protected technologies adjusts when a leader gains market access. This setting identifies adjustment when protection ends.

The remainder of the paper proceeds as follows. Section 2 describes the Import Source Diversification Program, its abolition, and the research design. Section 3 presents the data. Section 4 defines the bilateral quality measure and estimates the effects of liberalization on imports, domestic production and employment, and patenting in exposed technologies. Section 5 examines overall firm patenting, portfolio heterogeneity, and the technological destinations of reallocation.

2 Institutional Setting and Research Design

2.1 The Import Source Diversification Program

The Import Source Diversification Program (ISDP) was a country-specific non-tariff barrier in South Korea's postwar trade regime. Introduced in 1978, it sought to reduce Korea's persistent bilateral trade deficit with Japan and to protect domestic producers from Japanese import competition. Products on the ISDP list could not ordinarily be imported when their country of origin was Japan. Because the restriction applied to origin rather than shipment, Japanese goods routed through third countries remained covered. Ministerial approval was available only when an importer could demonstrate that the product could not feasibly be sourced outside Japan.

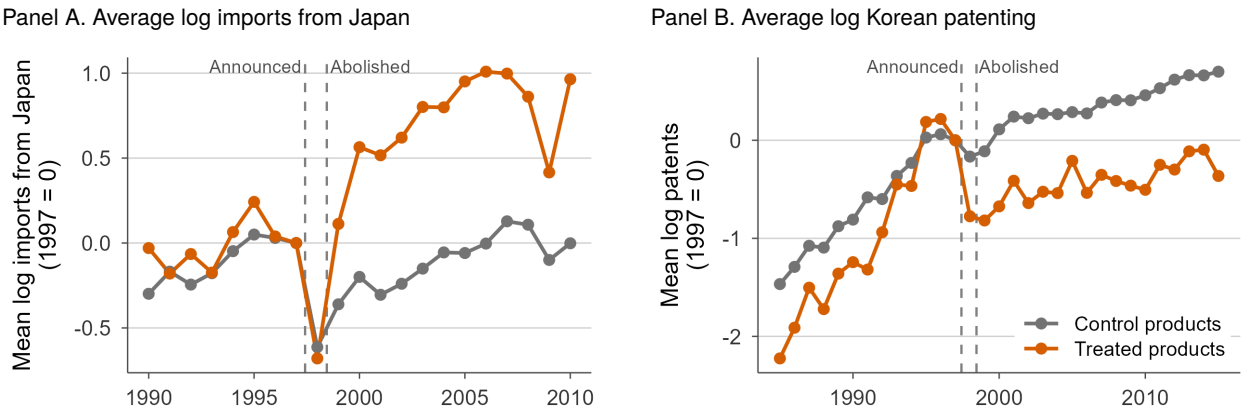


FIGURE 1. DESCRIPTIVE TRAJECTORIES OF TREATED AND CONTROL PRODUCTS

Notes: The panels compare average log imports from Japan and product-linked Korean corporate patenting, normalized to zero in 1997. Imports rise and patenting retreats in the exposed products following liberalization. Both panels use unweighted HS6 averages; patents are linked through Lybbert and Zolas (2014). Controls are all other products in each dataset. Logs use positive outcomes, and import observations require recorded world imports. There are 43 treated products in panel (A) and 42 with a patent concordance in panel (B). Vertical dashed lines mark the announcement and the first abolition wave.

The program overlaid Korea's otherwise MFN-based tariff regime with an additional restriction directed specifically at Japanese goods. It covered consumer products, capital equipment, and intermediate inputs, including automobiles, electronics, industrial machinery, and chemicals. Products outside the ISDP remained subject to Korea's ordinary MFN tariffs, with no bilateral preferential trade agreement in place.[1] The restriction was economically consequential because Japan accounted for approximately 20 to 30 percent of Korea's total imports during the 1990s.

The government gradually narrowed the ISDP during the 1990s. Following the Asian Financial Crisis, Korea announced the abolition of the remaining restrictions on December 3, 1997, and incorporated the commitment into the IMF agreement finalized on December 15. The remaining products were liberalized in two waves: one group on December 31, 1998, and the remainder on June 30, 1999.

The final-wave products were the residual set that had remained protected through the earlier phase-out, so assignment to this group was not random. What changed in 1997 was the timing of removal. The final timetable was determined under external IMF and WTO pressure. The empirical design exploits this externally determined timing while accounting for historical differences between final-wave and comparison products.

[1]According to UNCTAD TRAINS data, Korea applied a roughly uniform 8 percent tariff to most Japanese imports. Of the 43 final-wave HS6 products, 37 faced an 8 percent MFN tariff in 1997. The others faced weakly lower rates: steel fishing vessels 0 percent, small gasoline engines 4.0 percent, excavators 7.1 percent, and marine engine parts 7.6 percent. No 1997 rate is reported for wheat flour or for passenger cars above 2,500 cc, which faced 5 and 10 percent respectively in 1996.

Figure 1 previews the main empirical patterns. Treated and control products move broadly together before the abolition window. After 1998, imports from Japan rise sharply for treated products, while Korean patenting associated with those products declines relative to controls. The event-study specifications below examine these patterns within narrow product categories and conditional on predetermined characteristics.

2.2 Empirical Design

Treatment and timing. The policy shock is the abolition of restrictions on the final 43 HS6 products, spanning fourteen HS chapters and 36 HS4 groups. The original restrictions applied at the HSK10 level or below; affected lines represented, on average, 72.6 percent of their parent HS6 products' 1997 imports. Because the first abolition wave took effect on December 31, 1998, Japanese imports remained restricted throughout 1998. The import specifications therefore use 1998 as the reference year and trace effects beginning in 1999. Domestic firms could adjust their research and production plans after the December 1997 announcement, motivating 1997 as the reference year for patenting and establishment outcomes.

TABLE 1—THE FINAL ISDP LIST: 43 PRODUCTS ACROSS HS CHAPTERS

HS chapter	HS4 groups	Products	Representative products
Machinery (84)	13	14	Machining centers, NC lathes and mills, engines, forklifts, excavators, ball bearings
Electronics (85)	8	9	Mobile telephones, color televisions, camcorders, rice cookers, VCRs, generator sets
Vehicles (87)	2	6	Passenger cars in five engine and body classes, motor-vehicle parts
Other chapters	13	14	Refractory ceramics, 35 mm cameras, copiers, wheat flour, radial tires, watches, zippers
Total	**36**	**43**	

Notes: The table summarizes the final products liberalized under the ISDP. Machinery, electrical equipment, and vehicles account for most of the list: 29 of 43 products. The list spans 14 HS2 chapters and 36 HS4 groups. The remaining 14 products belong to 11 other HS2 chapters (11, 29, 39, 40, 55, 56, 69, 89, 90, 91, and 96), most represented by a single HS6 product. Under the BEC classification, 30 are final goods (consumer or capital goods) and 13 are intermediate goods. X-ray contrast media (HS 3006301000), which recorded zero imports from Japan in 18 of the 21 sample years, was excluded.

From products to technologies and firms. Section 4 links patent filings to products with the probabilistic IPC–HS concordance of Lybbert and Zolas (2014). Products with identical technology weights within an HS4 group are combined into a patent group. This permits a comparison of innovation associated with exposed and unexposed products.

Section 5 follows firms across narrowly defined CPC technologies to distinguish changes in their overall patenting from shifts in research direction. I combine sparsely populated technologies into cells with sustained patenting activity and identify those linked to the liberalized products using product descriptions and KIPO product–technology links. This classification tracks exposed and alternative technologies within the same firm's portfolio; Appendix C.2 describes the construction.

Estimation. For the product and establishment analyses in Section 4, I estimate event studies of the form

$$Y_{it} = \alpha_i + \lambda_{g(i)t} + \sum_{s \neq \tau_0} \beta_s D_i \mathbf{1}\{t = s\} + \sum_s \mathbf{1}\{t = s\} X_i' \Gamma_s + \varepsilon_{it}, \tag{1}$$

where D_i indicates treatment, τ_0 is the reference year, and X_i contains predetermined characteristics. Unit fixed effects control for permanent differences across products, technologies, or establishments, while category-by-year effects absorb shared shocks at the HS4 level for imports, the HS2 and IPC-class levels for patents, and the four-digit KSIC level for establishments. To account for trends associated with initial competitiveness and industry structure, the trade and patent specifications also interact year indicators with pre-liberalization Korean and Japanese revealed comparative advantage, domestic production concentration, and log production.[2]

For the firm-level analyses in Section 5, I use Poisson pseudo-maximum likelihood (PPML), which accommodates zero patent counts and estimates proportional responses. Because a few large patenters account for a substantial share of initial activity, I weight observations by the inverse of firms' pre-period patent counts.[3] These analyses trace both total patenting across firms and its composition within firms. Figures report 90 percent confidence intervals unless indicated otherwise.

2.3 Identification Strategy

Identification relies on the timing of the final abolition and comparisons with untreated products after accounting for category-by-year shocks and pre-treatment differences.

[2]To preserve the sample when control variables or quality measures are unavailable, I include indicators for these groups interacted with year effects, or with the post-period indicator in two-period specifications.

[3]If untreated expected patenting is proportional to initial patenting, this weighting yields $\log \mathbb{E}[e^{\delta_i}]$ in a binary-treatment benchmark, where δ_i is the firm's log effect. Appendix E.3 gives the derivation and compares alternative weighting schemes.

Conditional parallel trends. The event-study coefficients admit a causal interpretation if, absent the abolition, treated and comparison units in the same category $g(i)$ and with the same predetermined characteristics X_i would have followed parallel paths:

$$\mathbb{E}\left[Y_{it}(0) - Y_{i\tau_0}(0) \mid D_i = 1, g(i), X_i\right] = \mathbb{E}\left[Y_{it}(0) - Y_{i\tau_0}(0) \mid D_i = 0, g(i), X_i\right]. \tag{2}$$

Two features of the setting warrant particular attention: historical selection into the final-wave list and the coincidence of the announcement with the Asian Financial Crisis.

Selection into the final-wave group. The products that remained protected until the end may have differed systematically from products removed earlier or never covered. Unit fixed effects absorb time-invariant differences. Product- and technology-category-by-year fixed effects absorb shocks shared by related products or technologies, while interactions with predetermined characteristics allow products with different initial comparative advantage, scale, and concentration to follow separate annual paths. Pre-treatment paths provide a diagnostic for remaining divergence. Appendix A.1 describes initial differences; Appendix A.2 assesses alternative ways of constructing the comparison.

Technology-specific trends. Declining technological opportunities could reduce patenting in exposed technologies even without a competitive shock. Filings by public bodies, research institutes, and individuals provide a placebo: these applicants work in the same technologies but are less directly exposed to commercial competition. If common changes in technological opportunities drove the patent decline, filings by these less commercially exposed applicants might also fall (Appendix A.4).

The Asian Financial Crisis. The December 1997 announcement coincided with a severe contraction whose effects varied with leverage, domestic demand, export exposure, and firm structure. The sustained decline in patenting is less consistent with a predominantly crisis-driven response, which would predict a sharp fall in 1998 followed by recovery. Category-by-year fixed effects absorb common crisis shocks, while interactions with predetermined characteristics allow for heterogeneous adjustment. Appendix A.3 additionally allows annual responses to initial leverage.

Two additional comparisons help separate the competitive shock from the crisis. The source-country triple difference absorbs product-specific annual shocks, including changes in Korean demand common to

all sources. The within-firm patent specification absorbs firm-by-year shocks and asks whether firms shift patenting away from exposed technologies. The remaining identifying assumption is that the crisis did not cause differential changes in treated products or technologies beyond those captured by these controls.

3 Data

I combine bilateral trade flows, detailed US import data, Korean patent applications, and manufacturing survey records. Table 2 describes treated products and controls in the same HS4 categories.

TABLE 2—SUMMARY STATISTICS FOR TREATED AND CONTROL SAMPLES

	Treated			Control		
Variable	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.
Panel A: Trade Data (HS6 Products, 1990–2010)						
Imports from Japan ($ million)	903	33.0	58.1	3,882	11.7	41.3
Imports from world ($ million)	903	119.8	226.7	3,882	40.7	134.1
Korean RCA (1997)	43	1.08	1.17	188	0.68	1.62
Japanese RCA (1997)	43	2.30	1.85	188	1.52	1.83
Panel B: Patent Data (Patent Groups, 1985–2015)						
Patent count	1,178	108.9	875.0	4,278	42.2	198.4
Panel C: MMS Product Data (HS6 Products, 1997)						
Production value (billion KRW)	43	828.5	1,511.3	174	182.3	504.5
Number of establishments	43	60.9	179.1	174	29.0	92.8

Notes: The table compares pre-liberalization characteristics and outcomes of treated products with controls in the same HS4 categories. Treated products have larger average imports, patenting, and domestic production, motivating controls for initial scale and competitiveness. Entries report observation counts, means, and standard deviations. Import and patent observations are unit-by-year cells; RCA and MMS observations are one per product in 1997. Controls in each panel share an HS4 category with at least one treated product or patent group. Panel A uses product-years with positive world imports. Panel B includes zero patent counts in the descriptive statistics. Panel C reports 1997 MMS aggregates mapped to HS6 products. RCA follows Balassa (1965).

3.1 Trade data

I use bilateral trade flows from the Korea International Trade Association (KITA), harmonized to the HS 1996 six-digit classification. Descriptive series cover 1990–2010; event studies use 1990–2007. The main outcome is Korean imports from Japan. To avoid artificial changes caused by classification splits, I combine HS1996 products descending from the same HS1988/92 code into *trade groups*. A trade group is treated if any constituent product was on the final ISDP list.

The 43 treated products comprise 12 consumer goods, 18 capital goods, and 13 intermediate goods under the Broad Economic Categories (BEC) classification. Table 1 lists representative products by HS chapter.

To measure relative product quality, I use product-level US import data compiled by Schott (2008) from the US Census Bureau. The data report annual customs values and quantities by ten-digit HTS product and country of origin. I compare Japanese and Korean exports of the same HTS10 products to the United States, pooling 1996–1997 values and quantities to obtain pre-liberalization unit values. For treated HS6 categories, I select HTS10 subcategories that overlap with the scope of Korea's restrictions.[4] Section 4.4 defines the quality measure.

I also construct pre-treatment product characteristics used as controls in the event-study specifications. Korean and Japanese revealed comparative advantage are computed from 1997 UN Comtrade export data using the standard Balassa (1965) index. Domestic production concentration and pre-treatment production value are computed from product-level tabulations of the Mining and Manufacturing Survey. Group-level RCA uses aggregated exports; production concentration is weighted by constituent products' production values.

3.2 Patent Data

Patent applications provide a proxy for firms' innovation efforts and identify the technologies toward which those efforts are directed. R&D expenditure measures the resources devoted to research; patent classifications additionally reveal where firms conduct that activity. This technological detail makes patents suited to studying changes in the direction of innovation.[5]

I use KIPRIS records of applications filed with the Korean Intellectual Property Office during 1985–2015. To construct longitudinal firm patent portfolios, I harmonize corporate applicant identifiers by consolidating codes belonging to the same legal entity and linking documented business successors. This follows inventive activity through corporate restructuring, which could otherwise appear as a firm's disappearance from the patent record. I exclude corporate histories that became wholly foreign-owned during 1993–2007. Appendix C.1 describes the linkage criteria, and Table C1 summarizes sample construction. Firm characteristics use filings through 1997; the firm event studies cover 1993–2007.

[4] Quantities use the first statistical unit reported for each product, with positive values and quantities required for both countries. The sample includes 5,625 common HTS10 products across 2,498 HS6 categories; 178 detailed products cover 39 of the 43 treated HS6 categories.

[5] Counting each application equally abstracts from differences in its importance. Appendix A.5 weights patents using text-based importance measures following Kelly et al. (2021).

For the destination analysis, I use Google Patents BigQuery publication records from the US, European, Japanese, and Chinese patent offices to measure pre-liberalization foreign patent growth and stocks (Appendix E.7).

Technological classifications. The Cooperative Patent Classification (CPC) describes technologies hierarchically, from broad sections to detailed subgroups. Because an application may cover several technologies, I divide its count equally across its recorded CPC symbols and sum these contributions within CPC cells to measure technological portfolios. Total firm patenting counts unique applications. The IPC three-character *class* groups technologies by shared functions or technical principles; the finer four-character *subclass*, harmonized across classification revisions, provides the categories used for product links and destination characteristics.

I map IPC codes to the 35 technological *fields* and five broad *sectors* of the WIPO IPC–Technology Concordance Table (Schmoch, 2008). These fields describe areas of technological application. Fields containing a firm's treated pre-period patents define its affected fields; experience in the remaining fields measures its knowledge base in alternative applications.

Product-linked patent counts. Applying the Lybbert and Zolas (2014) IPC–HS weights to patent counts produces the product-linked series in Section 4. Patent groups sharing a weight vector, including groups in different HS4 categories, form the concordance clusters used for standard errors. The analysis focuses on technologies with sustained patenting before and after liberalization.[6]

3.3 Mining and Manufacturing Survey

I measure domestic production using the Korean Mining and Manufacturing Survey, a near-census of manufacturing establishments with five or more employees. The data cover 1991–2007 and report establishment-level production, employment, and product codes at the eight-digit level, consisting of a five-digit KSIC industry prefix and a three-digit product sequence.

To link product-level liberalization to establishment outcomes, I match the survey's product codes to HS6 products using a hand-constructed concordance. An establishment is treated if, in at least one year during 1991–1997, it reports a single product mapped to the final ISDP list.

[6]Log specifications use positive annual counts for patent groups with filings in at least five years in 1985–1997 and five years in 1998–2015.

The 1997 product aggregates also measure initial domestic production and concentration. Together, the trade, establishment, and patent records trace how the removal of protection changed market competition, domestic activity, and the direction of research.

4 Import Competition, Production, and Innovation

The abolition first changed the competitive environment facing Korean producers. I establish the increase in Japanese imports, trace the response of domestic production and employment, and then examine patenting in the exposed technologies. The initial Japan–Korea quality gap provides a measure of the foreign advantage against which domestic firms had to compete.

4.1 Japanese Import Competition

Japanese imports of the liberalized products rose by approximately 66 percent and remained above their pre-liberalization path (Figure 2, panel A). The comparison follows exposed and unexposed trade groups within the same narrow product categories, accounting for their initial competitiveness and domestic production structure.

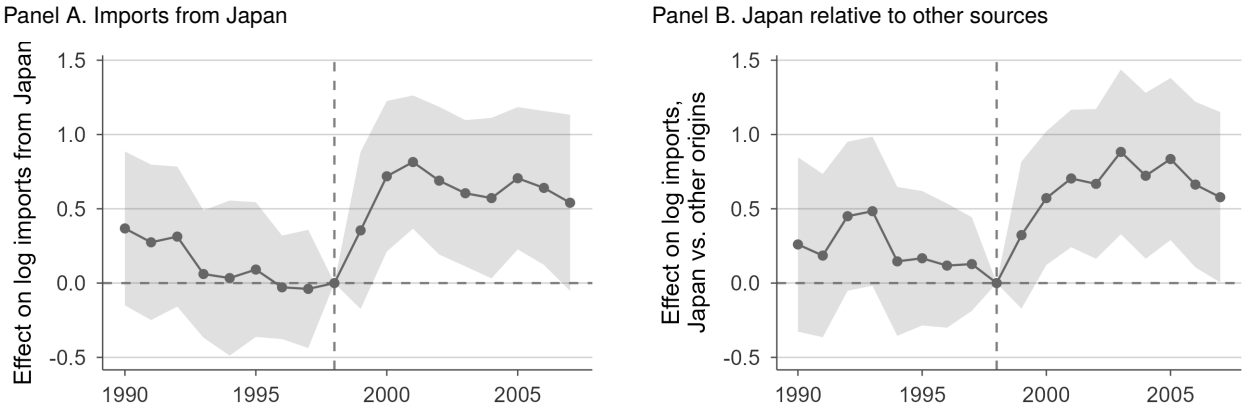


FIGURE 2. EFFECT OF ISDP ABOLITION ON JAPANESE IMPORT COMPETITION

Notes: The panels estimate the import response to ISDP abolition, comparing exposed with unexposed trade groups in panel (A) and Japan with other sources in panel (B). Both comparisons show a persistent increase in Japanese imports. Pooled treatment-by-post coefficients (clustered standard errors in parentheses) are 0.507 (0.221) and 0.447 (0.233), respectively. Panel (A) includes group and HS4-by-year effects and year interactions with initial RCA, concentration, and production; panel (B) follows equation (3). Observations: 54,628 and 603,929; trade-group clusters: 3,142 and 4,142. The sample covers 1990–2007, with 1998 as the reference year and at least five positive-import years before and after liberalization. Bands are 90 percent confidence intervals.

Panel (B) compares Japan with other source countries:

$$\log M_{ict} = \alpha_{ic} + \lambda_{it} + \mu_{ct} + \sum_{s \neq 1998} \delta_s D_i \mathbf{1}\{c = J\}\mathbf{1}\{t = s\} + \varepsilon_{ict}. \tag{3}$$

Trade-group-by-year effects absorb changes in Korean demand common to all sources, and source-by-year effects absorb source-wide shocks. Japanese imports rise by approximately 56 percent relative to other sources. Together, the two comparisons identify a persistent increase in Japanese competition following the removal of the country-specific restriction.

A surge in Japanese imports could displace goods from other foreign suppliers or reduce domestic production. The response of production and employment reveals how this competitive shock affected Korean manufacturers.

4.2 Domestic Production and Employment

Korean manufacturers faced a loss of protection in their home market. I trace the domestic response using establishment-level production and employment from the Mining and Manufacturing Survey. The analysis focuses on final goods because Japanese subsidiaries, which the survey does not separately identify, were concentrated in intermediate-goods manufacturing (Appendix D.2). Establishment and industry-by-year effects account for persistent differences and common industry shocks, while initial capital, employment, and product scope allow for different adjustment paths.

Treated final-goods and control establishments have similar pre-liberalization production and employment distributions (Table 3). This comparison provides a basis for tracing their subsequent divergence within the same industries.

Production and employment declined gradually among single-product establishments (Figure 3). By 2007, production was approximately 30 percent lower. Averaging the annual log responses over the post-liberalization period implies reductions of about 18 percent in production and 8 percent in employment. Multi-product establishments also contracted, with smaller production losses consistent with greater scope to adjust across product lines (Appendix Figure D1).

The withdrawal of protection therefore reduced domestic production value and employment over the subsequent decade. Lower product-market returns can change incentives to innovate: competition may encourage firms to catch up with foreign rivals, or discourage research by reducing the rewards to invention.

Table 3—Pre-Treatment Characteristics of Establishments, 1997

	Control			Final-goods treated		
Establishments	62,571			390		
Multi-product establishments	20,002			100		
	Q25	Median	Q75	Q25	Median	Q75
Production value	290	687	1,980	299	702	1,897
Labor	7	12	25	7	12	26
Labor share	0.125	0.197	0.309	0.138	0.210	0.297
Capital	57	228	851	87	286	754

Notes: The table compares the pre-liberalization distributions of production, labor, and capital among treated and control establishments. Production and employment distributions are similar across the groups. Establishments are observed in both 1997 and 1998. Treatment uses pre-1998 production of a single reported product mapped to a treated final good. Multi-product status indicates at least two reported products in 1997 or 1998. Production and capital are in millions of KRW; labor counts employees, while the employment regressions also include the owner. Labor share is labor cost divided by production value. The table reports the all-industry balance sample; industry-by-year fixed effects and outcome availability determine the estimation sample.

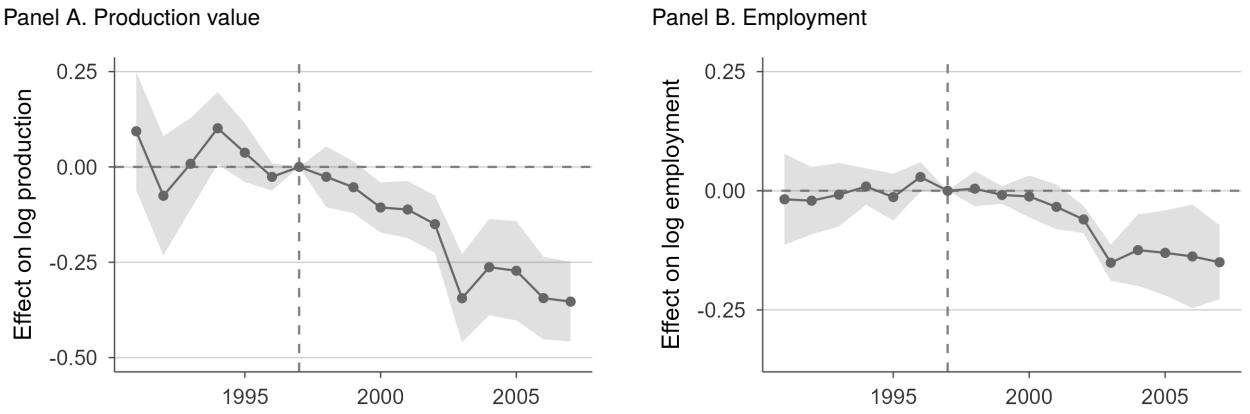


Figure 3. Production and Employment in Single-Product Establishments

Notes: The panels estimate production and employment responses among single-product final-goods establishments. Domestic activity contracts gradually after liberalization. Mean annual post-period coefficients are -0.202 (0.037) for production and -0.080 for employment. Both regressions include establishment and KSIC4-by-year effects, year interactions with initial capital, employment, and multi-product status, and joint final- and intermediate-goods treatment interactions. The sample covers 1991–2007, relative to 1997, with 668,319 observations and 64,502 establishments. Bands are 90 percent confidence intervals clustered by four-digit industry.

4.3 Patenting in Exposed Technologies

Domestic firms could respond to stronger foreign competition by intensifying research or by retreating from the exposed technologies. To assess which response dominated, I compare corporate patent filings in exposed and unexposed patent groups (Figure 4), using the product concordance described in Section 3.2. The specification follows equation (1), allowing innovation to respond from the announcement of liberalization.

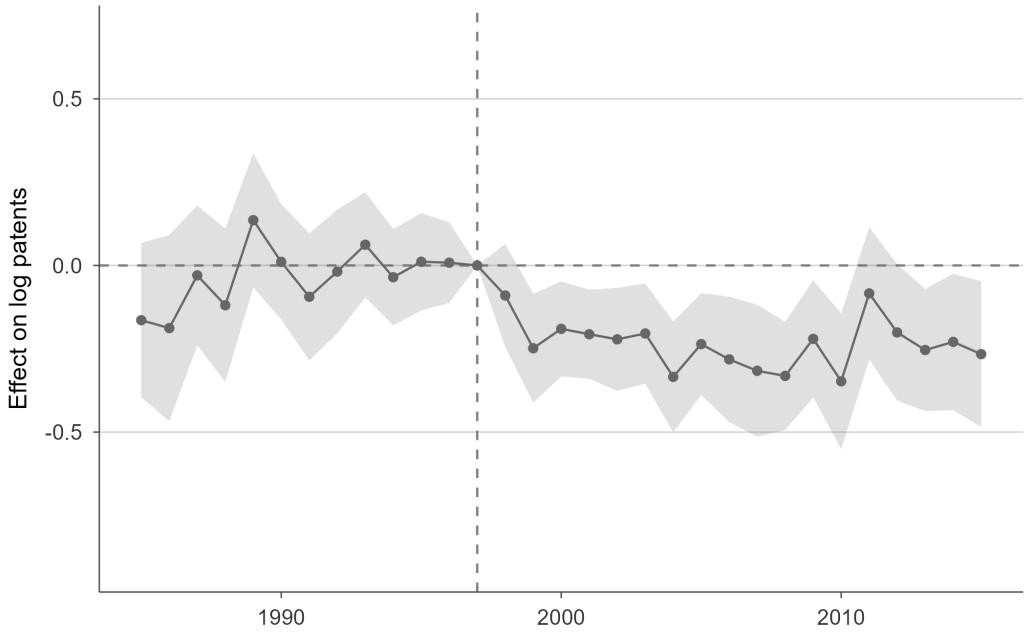


FIGURE 4. EFFECT OF ISDP ABOLITION ON KOREAN CORPORATE PATENT FILINGS

Notes: The figure estimates the response of Korean corporate patenting in product-linked technology groups. Patenting falls soon after the announcement and remains below its comparison path. The pooled treatment-by-post coefficient is -0.206 (0.088). The sample covers 1985–2015, relative to 1997, with 107,022 observations, 3,612 patent groups (36 treated), and 3,492 concordance clusters. Patent-group, HS2-by-year, and IPC-class-by-year effects and year interactions with initial RCA, production concentration, and log production follow equation (1). Bands are 90 percent confidence intervals clustered by concordance group; sample eligibility is described in Section 3.2.

Patenting fell by approximately 19 percent, with the decline emerging soon after the announcement and persisting through the end of the sample. The retreat also appears with alternative comparison methods, importance-weighted patents, and the independent CPC classification (Appendices A.2, A.5, and C.3).

If common changes in technological opportunities drove the patent decline, filings by less commercially exposed applicants might also fall. Public bodies, research institutes, and individuals show a smaller response, supporting an interpretation tied to commercial competition (Appendix A.4).

Patenting adjusted before the largest production losses materialized, consistent with firms reducing research in anticipation of lower returns. The strength of this incentive should depend on how likely domestic firms are to catch up with the foreign competitor, motivating an examination of their initial technological distance.

4.4 Patent Responses by the Initial Quality Gap

When domestic firms face a technological leader, their return to continued research depends on the prospect of closing the initial gap (Aghion et al., 2005, 2009; Akcigit et al., 2026). A larger foreign quality lead makes catch-up less likely and may strengthen the incentive to retreat. I test this prediction using Japan's pre-liberalization quality advantage in the affected products.

Measuring the bilateral quality gap. For HTS10 product h and origin $c \in \{J, K\}$, let V_h^c and Q_h^c denote the value and quantity of US imports pooled over 1996–1997, and let $P_h^c = V_h^c/Q_h^c$. Following Khandelwal et al. (2013), CES demand implies $V_h^c = A_h(\lambda_h^c/P_h^c)^{\sigma-1}$, where A_h is common to the two origins and λ_h^c is demand-inferred quality. The bilateral log quality gap is

$$q_h = \log \frac{\lambda_h^J}{\lambda_h^K} = \log \frac{P_h^J}{P_h^K} + \frac{1}{\sigma - 1} \log \frac{V_h^J}{V_h^K}, \qquad \sigma = 5. \tag{4}$$

At a given relative price, larger Japanese sales imply higher Japanese quality; a positive gap denotes a Japanese advantage. The common US market permits this comparison without using Korean imports restricted by the ISDP.

I aggregate the HTS10 gaps to HS6 products using weights proportional to the two countries' combined sales, $V_h^J + V_h^K$. For treated products, I use the fully or partially covered HTS10 products described in Section 3.1; for other products, I use all eligible lines. Quality is observed for 39 of the 43 treated products.[7]

Japan has a quality advantage in 35 of the 39 observed treated products, with substantial variation in its size (Appendix Figure B1). The gap is positively correlated with independent measures of technological standing, revealed comparative advantage, and import penetration (Appendix Figure B2). These relationships connect the quality gap to independent dimensions of bilateral competitive standing.

[7]The four missing products are HS 550200, 844630, 870333, and 890200. A treated regression group's gap is the simple average across its constituent treated products with observed quality.

Patenting falls by approximately 26 percent in technologies linked to products with an above-median quality gap, compared with 14 percent below the median (Figure 5). The larger decline where Japan held a greater initial lead connects the innovation response to the difficulty of competing with the foreign frontier.

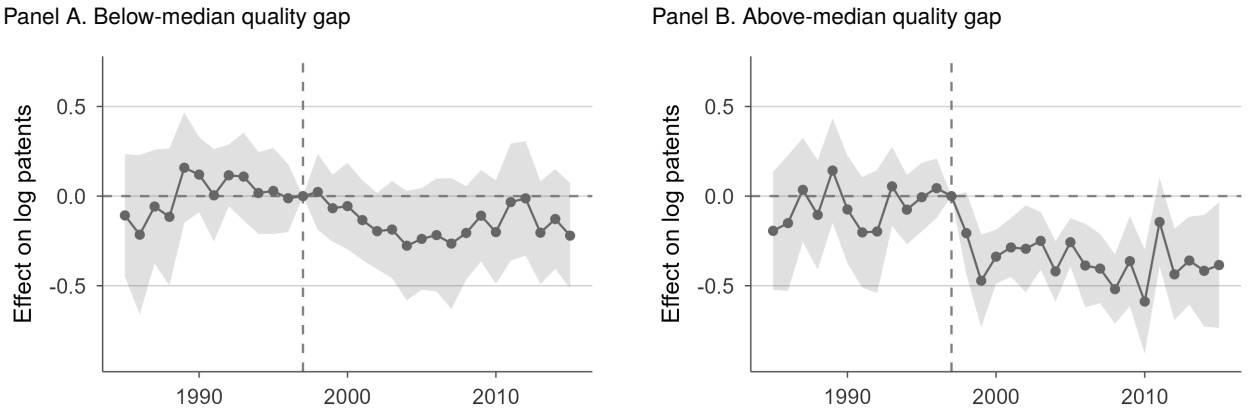


FIGURE 5. PATENT RESPONSES BY THE INITIAL JAPAN–KOREA QUALITY GAP

Notes: The panels jointly estimate patent responses below and above the median initial Japan–Korea quality gap. The decline is larger in technologies linked to products with a greater Japanese quality lead. Pooled treatment-by-post coefficients (clustered standard errors in parentheses) are -0.155 (0.166) and -0.307 (0.078), respectively. The cutoff is 1.238; the estimation sample includes 17 below-median and 16 above-median treated groups with measured quality, 107,023 observations, 3,612 patent groups, and 3,492 concordance clusters. The specification follows Figure 4. Bands are 90 percent confidence intervals relative to 1997.

The independent CPC classification yields larger declines in high-gap technologies under each treatment definition, and a patent-based bilateral gap gives the same ordering (Appendices C.3 and B.3). The quality gradient is weaker among non-corporate applicants (Appendix Figure A6B). Imports also rise more in products with a larger initial Japanese advantage (Appendix B.4), linking the patent gradient to both the intensity of foreign competition and the initial difficulty of catch-up.

Together, the timing and heterogeneity of the technology-level responses point to a decline in expected research returns in the exposed fields. Patenting adjusts before the main contraction in domestic production and falls most where Japan initially held a larger quality lead. This retreat can involve a reduction in firms' overall innovation or a redirection of research toward other technologies.

Technology-level aggregates combine these two margins. Firm portfolios reveal whether incumbents contract their overall inventive activity, as a broad loss of financial capacity might induce, or sustain patenting while shifting its technological direction.

5 Reallocation of Innovation within Firms

To separate an overall contraction in corporate innovation from a shift in technological direction, I trace both the total patent filings of exposed incumbents and the composition of their portfolios. I then examine how prior knowledge in alternative fields shapes adjustment and where inventive activity moves.

5.1 Total Patenting and Within-Firm Reallocation

Exposed incumbents may reduce their total research activity or redirect it as returns fall in protected technologies. I first compare their total patent applications with those of firms with little exposure and similar initial research activity. Exposure is the share of a firm's pre-liberalization patent counts in treated CPC cells. I match firms whose main fields belong to the same sector, using prior filing activity and patenting volume to align their initial scale and IPC-class portfolio similarity to align their technological knowledge bases.[8]

Let N_{it} be firm i's annual number of unique applications, E_i its exposure, and D_i an indicator for the exposed member of a matched set. I estimate

$$\mathbb{E}[N_{it} \mid \cdot] = \exp\left\{\alpha_i + \lambda_{m(i)t} + \sum_{s \neq 1997} \beta_s D_i \frac{E_i}{\bar{E}} \mathbf{1}\{t = s\} + X_i' \rho_t\right\}, \tag{5}$$

where $\lambda_{m(i)t}$ denotes matched-set-by-year effects and X_i allows trends to vary with initial patenting. Exposure is scaled by the median among matched exposed firms, so the plotted responses describe a firm at that reference exposure. Total patenting changes little on average (Figure 6, panel A).

Firms that sustain their total inventive activity can still withdraw from particular technologies. I therefore compare exposed and unexposed patenting within each firm. For patent counts P_{ict} in CPC cell c, I estimate

$$\mathbb{E}[P_{ict} \mid \cdot] = \exp\left\{\alpha_{ic} + \lambda_{it} + \eta_{g(c)t} + \sum_{s \neq 1997} \theta_s T_c \frac{E_i}{\bar{E}} \mathbf{1}\{t = s\}\right\}, \tag{6}$$

where T_c identifies treated cells and $g(c)$ denotes the IPC class. Firm-by-year effects absorb changes in firms' overall patenting, while class-by-year effects remove common trends among technologies with related

[8] Exposed candidates have exposure of at least 5 percent, and controls at most 1 percent; both have a patent count of at least five during 1985–1997. Appendix E.1 describes the full matching procedure and balance.

functions or principles. The sample follows firms with pre-liberalization experience in both exposed and unexposed technologies.[9] The reference exposure is the same as in the between-firm comparison.

Within firms, patenting in treated technologies falls by about 29 percent relative to untreated technologies, based on the mean annual post-liberalization response (Figure 6, panel B).[10]

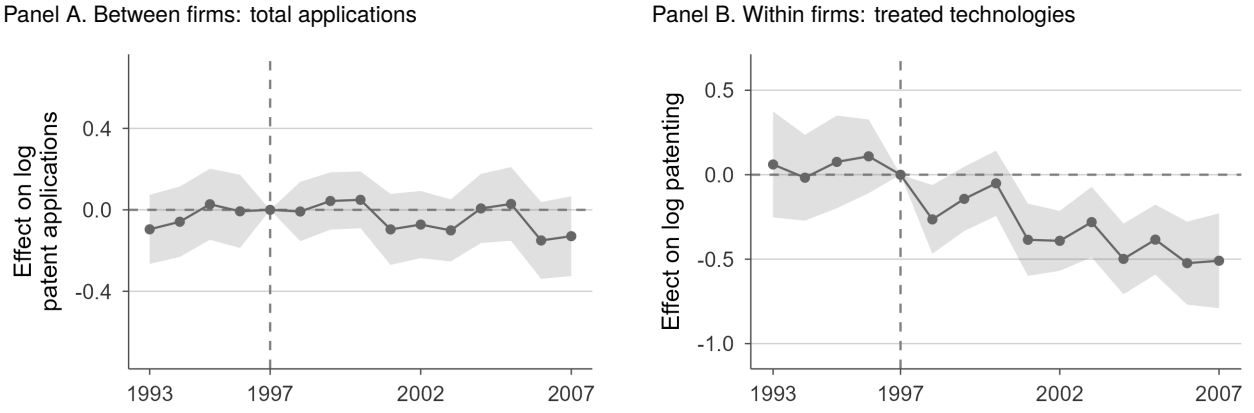


FIGURE 6. TOTAL PATENTING AND WITHIN-FIRM REALLOCATION

Notes: The panels estimate changes in total patenting between matched firms and in treated relative to untreated patenting within firms. Total patenting changes little on average while its composition shifts away from exposed technologies. Pooled treatment-by-post coefficients (firm-clustered standard errors in parentheses) are −0.019 (0.069) and −0.341 (0.073); mean annual post-period coefficients are −0.043 (0.085) and −0.343 (0.095). Equations (5) and (6) cover 1993–2007, relative to 1997, with inverse pre-period patent-count weights and a common reference exposure of 10.75 percent. Panel (A) also adjusts for reuse across matched sets and includes year interactions with log initial patenting separately by portfolio group. Observations: 11,323 and 100,955; firm clusters: 697 (238 exposed, 459 controls) and 504. Bands are 90 percent confidence intervals. Appendix E.1 gives sample requirements.

The average exposed incumbent thus changes the composition of its patenting while sustaining its overall scale. Redirecting research requires expertise in the destination, however, so this adjustment should depend on the knowledge firms had already accumulated outside the affected activities.

5.2 Prior Knowledge in Other Fields

Research builds cumulatively on existing expertise, making prior knowledge in alternative applications a potential source of flexibility. I define a firm's affected fields as the WIPO fields containing its treated pre-period patents. Untreated patents in these fields are *same-field* patents; patents in all remaining fields are

[9] Firms have exposure of at least 1 percent and, during 1985–1997, at least one application containing a treated cell and one containing none. Appendix E.1 gives the remaining eligibility conditions.

[10] At the reference exposure of 10.75 percent, a 3.2 percent increase in untreated patenting would offset this relative decline and keep total patenting constant. In the raw data, exposed firms' total patent growth has a distribution similar to that of all firms in the descriptive sample. Within exposed firms, patenting declines more in treated than in untreated technologies on average (Appendix Figure E3).

other-field patents. For a firm whose treated patents lie only in transport, for example, untreated transport patents belong to the same field, while optics patents belong to another field. This distinction is specific to each firm and fixed using its pre-liberalization portfolio.

I measure outside experience by the share of a firm's untreated pre-period patenting located in other fields. This share captures the extent of its knowledge base in alternative applications. Firms are divided at the median among matched exposed firms, using the same threshold for the between-firm and within-firm comparisons.

High-share firms do not exhibit the decline in total patenting observed among low-share firms (Figure 7).[11] Their annual estimates imply an increase of about 24 percent in total patenting alongside a relative decline of about 40 percent in treated technologies. Firms with a larger knowledge base in other fields therefore adjust the direction of their portfolios while sustaining overall inventive activity.

Low-share firms show less reallocation: treated patenting falls by about 15 percent relative to untreated patenting. Their total patenting also declines gradually, with an average annual response corresponding to a reduction of about 17 percent. This dynamic decline is significant at the 10 percent level.[12]

[11] These patterns are not driven by any single firm or treated product (Appendix Figures E8 and E9). The low-share group includes more firms with large initial patent portfolios, and the differences persist when such firms are excluded (Appendix E.5). The total-patenting heterogeneity also persists with year interactions for 1997 log employment, log revenue, leverage, and ROA in the ValueSearch-linked sample (Appendix Figure E2).

[12] Figure 7 reports the mean annual response and its uncertainty alongside the pooled treatment-by-post estimate. Restricting the sample to firms with a filing in 2007 or later preserves the relative retreat within both groups and the decline in total patenting among low-share firms (Appendix Figure E7).



FIGURE 7. INNOVATION RESPONSES BY PRIOR EXPERIENCE IN OTHER FIELDS

Notes: The panels estimate responses by prior experience in other fields, jointly within each row. Firms with more outside-field experience reallocate more strongly, while the gradual decline in total patenting occurs among low-share firms. High denotes an other-field share above 70.79 percent. In panel order, pooled treatment-by-post coefficients (firm-clustered standard errors in parentheses) are 0.083 (0.105), −0.084 (0.093), −0.541 (0.114), and −0.185 (0.081); mean annual post-period coefficients are 0.218 (0.133), −0.190 (0.100), −0.503 (0.155), and −0.164 (0.092). The top row includes 120 high-share and 118 low-share exposed firms, with 459 controls; the bottom row includes 253 high-share and 251 low-share firms. Matched controls follow their exposed firm's group. Observations and firm clusters are 11,323 and 697 in the top row, and 100,955 and 504 in the bottom row. Year interactions with initial patenting in the top row are group-specific. Other settings and 90 percent intervals follow Figure 6.

Prior knowledge in other fields thus provides an outside option when research returns fall in protected activities. The contrast suggests that firms with a broader base of alternative capabilities can redirect effort more readily. It also raises a question about the destinations: whether research shifts to neighboring technologies in the affected field or toward other fields with expanding opportunities.

5.3 Destinations of Reallocation

Reallocating firms can pursue adjacent, unexposed technologies within the affected field or move into different fields. If they seek expanding research opportunities, those destinations should also exhibit stronger technological growth. I examine the location of adjustment first, then relate destination characteristics to changes in portfolio shares. In equation (6), separate exposure-by-year interactions for same-field and other-field technologies use treated cells as the comparison, preserving the sample, weights, and fixed effects.

The larger relative gains occur outside the affected fields, consistent with the stronger reallocation by firms with prior knowledge in those destinations (Figure 8). The annual estimates imply that same-field patenting rises by about 16 percent relative to treated technologies, compared with 57 percent for other-field patenting. The relative expansion is therefore greater across fields than within the affected field.

This pattern is consistent with research activities in a field sharing a knowledge-production base. When competition reduces the return to some products built on that knowledge, firms can scale back research serving the broader field and redirect effort toward other fields. Firms with prior experience in those destinations have a knowledge base on which to build. The evidence in Figures 7 and 8 connects this interpretation to both the firm's initial portfolio and the direction of its subsequent adjustment.

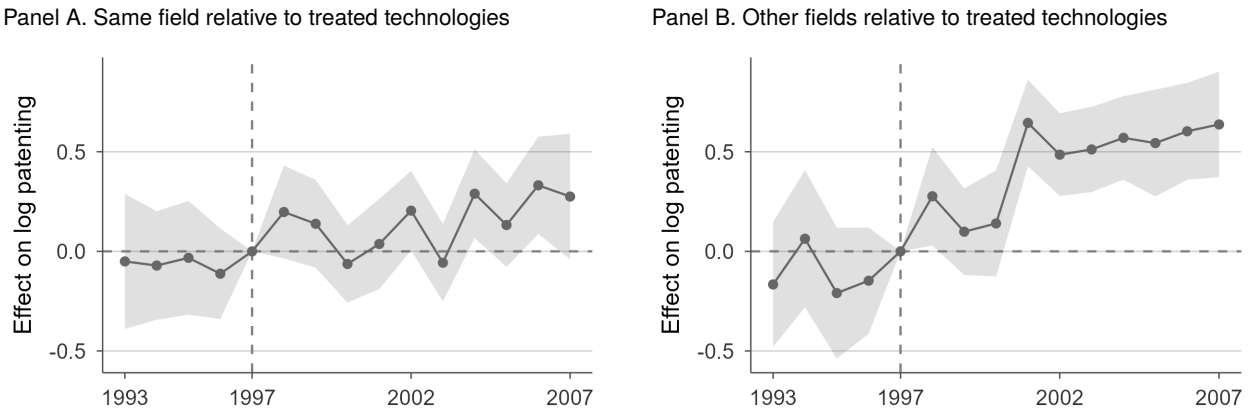


FIGURE 8. THE TECHNOLOGICAL DESTINATIONS OF REALLOCATION

Notes: The panels jointly estimate same-field and other-field patenting relative to treated technologies within firms. The larger relative expansion occurs in other fields. Pooled treatment-by-post coefficients (firm-clustered standard errors in parentheses) are 0.189 (0.084) and 0.495 (0.082); mean annual post-period coefficients are 0.149 (0.100) and 0.451 (0.117). Untreated cells belong to the same field if their WIPO field contains the firm's treated patents in 1985–1997. The model uses 106,379 observations and 504 firm clusters, with firm–cell, firm–year, and IPC-class–year effects. Weights, the 10.75 percent reference exposure, the 1997 reference year, and 90 percent intervals follow Figure 6(B).

To characterize the opportunities attracting this adjustment, I relate changes in firms' IPC-subclass portfolio shares between the ten years before and after liberalization to pre-liberalization technological characteristics. Regressions are estimated separately for exposed and low-exposure firms, conditioning on firm fixed effects and the initial portfolio share. The five characteristics measure relatedness to the firm's prior knowledge, domestic and foreign patent growth, and domestic and foreign patent stocks. Growth compares the two five-year windows before liberalization, while stocks measure cumulative patenting over that decade. Table 4 reports the regressions, and Appendix E.7 defines the variables and samples.

Firms expand their portfolio shares in technologies related to their existing knowledge. In column (6), a one-standard-deviation increase in prior foreign patent growth is associated with a 0.036-percentage-point increase in exposed firms' portfolio shares, over four times the 0.008-point increase among low-exposure firms. Technologies with rapid foreign growth include wireless communication networks (H04W), electric-vehicle propulsion (B60L), and digital information transmission (H04L); patent activity at the foreign offices grew by factors of approximately 3.8, 3.6, and 2.6 across the two five-year windows.

A complementary within-firm event study also yields a positive destination gradient in foreign growth (Appendix E.8). Taken together, the portfolio evidence suggests that exposed firms redirected inventive activity toward other fields and, within those destinations, toward technologies expanding abroad. This adjustment links the loss of protection to a change in research direction aligned with international technological opportunities.

TABLE 4—TECHNOLOGICAL CHARACTERISTICS AND CHANGES IN PATENT PORTFOLIO SHARES

Dependent variable	Change in IPC-subclass portfolio share (pp)					
Panel A. Exposed firms ($E_i \geq 5\%$)						
	(1)	(2)	(3)	(4)	(5)	(6)
Own relatedness	0.1095***					0.0983***
	(0.0181)					(0.0174)
Domestic growth		0.0254***				0.0106
		(0.0090)				(0.0088)
Domestic stock			0.1017***			0.0909***
			(0.0152)			(0.0280)
Foreign growth				0.0601***		0.0359**
				(0.0181)		(0.0166)
Foreign stock					0.0852***	-0.0211
					(0.0148)	(0.0266)
Observations	101,906	101,906	101,906	101,906	101,906	101,906
Firms	203	203	203	203	203	203
IPC subclasses	502	502	502	502	502	502
Within R-squared	0.1913	0.1890	0.1899	0.1893	0.1897	0.1922
Panel B. Low-exposure firms ($E_i \leq 1\%$)						
	(1)	(2)	(3)	(4)	(5)	(6)
Own relatedness	0.1004***					0.0922***
	(0.0087)					(0.0086)
Domestic growth		0.0101**				-0.0054
		(0.0044)				(0.0044)
Domestic stock			0.0868***			0.0756***
			(0.0100)			(0.0130)
Foreign growth				0.0250***		0.0083
				(0.0071)		(0.0059)
Foreign stock					0.0721***	-0.0066
					(0.0100)	(0.0144)
Observations	445,776	445,776	445,776	445,776	445,776	445,776
Firms	888	888	888	888	888	888
IPC subclasses	502	502	502	502	502	502
Within R-squared	0.1707	0.1691	0.1699	0.1692	0.1697	0.1712

Notes: The regressions relate changes in firms' IPC-subclass portfolio shares (percentage points) between 1988–1997 and 1998–2007 to destination characteristics. Exposed firms shift toward related technologies and, in the joint regression, toward those growing faster abroad. Columns (1)–(5) enter one characteristic each; column (6) enters all five. All include firm effects, the initial share, its square, and an indicator for an initially zero share. Characteristics use a common standardization across samples. Growth compares 1988–1992 with 1993–1997; stocks cover 1988–1997; relatedness uses the 1985–1997 portfolio. Firms have a pre-period patent count of at least five and positive patenting in both windows. Regressions are unweighted; parentheses report standard errors clustered by firm and IPC subclass. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

6 Conclusion

South Korea's final abolition of the Import Source Diversification Program generated a sharp, Japan-specific increase in imports and a sustained decline in Korean patenting in exposed technologies. The decline was larger where Japan held a greater pre-liberalization quality advantage and emerged before the largest losses in domestic production. Together, these patterns suggest that competition from a technological leader lowered the expected returns to innovation in exposed technologies, particularly where domestic producers faced greater obstacles to catching up.

The firm-level evidence shows that retreat from these technologies did not uniformly translate into less innovation overall. Incumbents with larger shares of prior patenting outside the affected fields maintained total patenting while shifting toward other fields. Firms whose prior patenting was concentrated in and around the exposed technologies reallocated less and experienced a gradual decline in total patenting. Across exposed firms, portfolio changes favored technologies in which patenting abroad had been growing faster before liberalization. These findings suggest that prior experience in other fields gave firms greater scope to pursue alternative technological opportunities when competition made continued innovation in the exposed fields less attractive.

This reallocation also explains why technology-level and firm-level estimates can convey different pictures of the innovation response to competition. Technology-level estimates track patenting within exposed technologies, whereas firm-level estimates of total patenting follow firms across the technologies in which they innovate. A firm's shift away from an exposed technology can therefore contribute to a technology-level decline without reducing its total patenting. The estimates capture distinct margins of response and should be interpreted accordingly: declining patenting in exposed technologies need not imply a comparable contraction in exposed firms' total patenting, while stable total patenting can conceal substantial changes in the technologies those firms develop.

For industrial policy, this distinction separates the persistence of domestic innovation from the persistence of the technological specialization that protection supported. After protection ends, firms may sustain innovation by moving beyond the technologies the policy sought to develop. The consequences of withdrawal thus depend partly on the innovation histories firms bring to liberalization, including their experience outside the protected fields. Evaluating those consequences requires following both the targeted technologies and the firms that developed them. The capacity to sustain domestic innovation under foreign competition may rest partly on firms' ability to innovate beyond the original policy targets.

Appendix

Appendix A assesses policy selection, alternative comparisons, financial conditions, and patent measurement. Appendix B documents the bilateral quality measure and its relationship to patenting and imports. Appendix C describes applicant histories, technology cells, and direct product–technology links. Appendix D examines establishment responses by product scope and the role of Japanese subsidiaries. Appendix E explains matching and weighting, summarizes the robustness of firm responses, and develops the destination analysis.

A Identification and Robustness of Product-Level Results

A.1 Policy Selection and Pre-Treatment Characteristics

The restrictions selected products rather than assigning treatment randomly. Figure A1 compares the pre-liberalization distributions of Korean and Japanese comparative advantage, production concentration, and domestic production. These characteristics enter the baseline regressions interacted with year, alongside narrowly defined product-category effects. The event studies assess whether exposed and comparison technologies had already followed different paths before the policy change.

A.2 Alternative Comparisons and Inference

Differences in initial trends or covariates could affect the comparison path. Synthetic difference-in-differences reweights comparison groups and pre-period years to improve the pre-treatment fit, while doubly robust difference-in-differences adjusts for observed characteristics through outcome and treatment models (Figure A2). Both approaches preserve the patent decline and its concentration in the high-gap group, supporting the interpretation of the baseline response.

Figure A3 reports randomization-based intervals from 999 reassigned treatment sets of the observed size. The exercise assesses inference with a relatively small number of treated groups while holding the baseline specifications fixed. Quality groups are reassigned jointly with treatment, preserving their sizes.

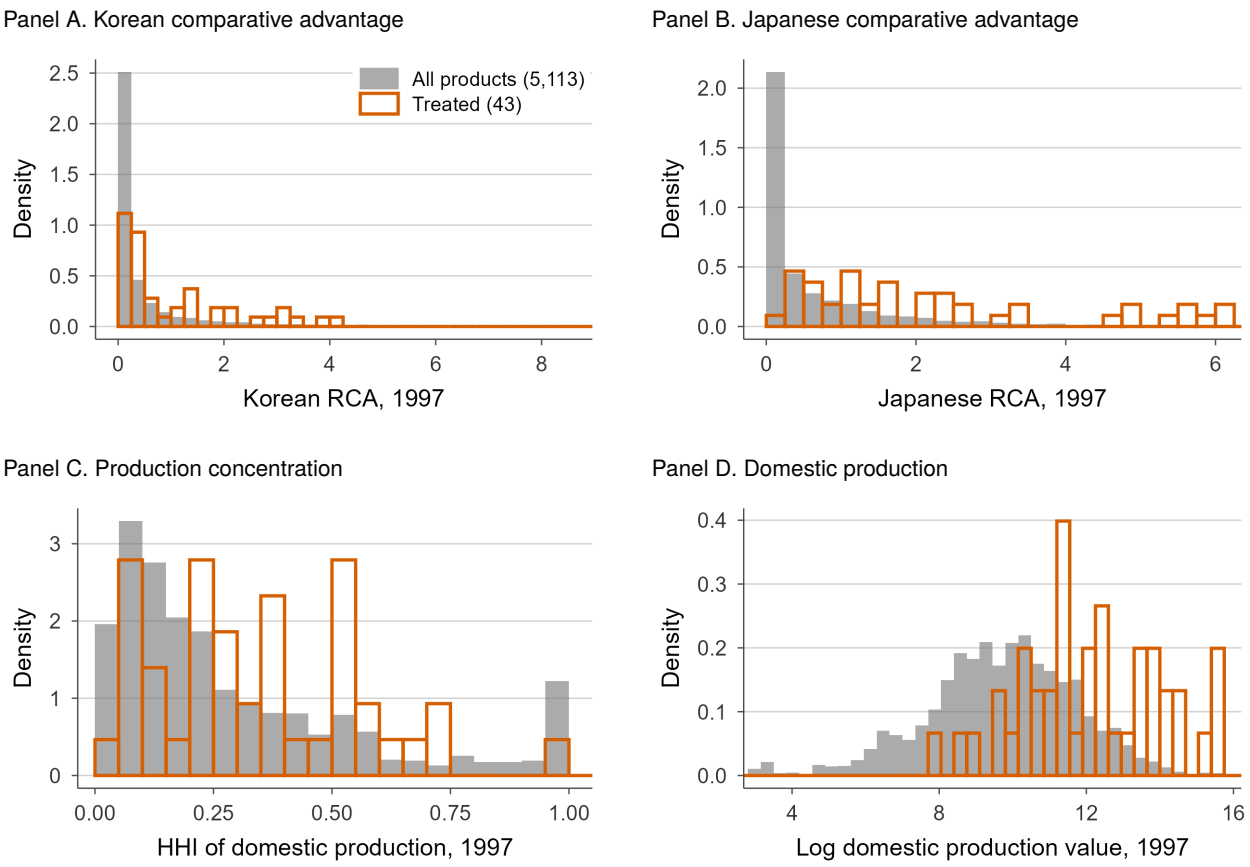


FIGURE A1. PRE-LIBERALIZATION CHARACTERISTICS OF TREATED PRODUCTS AND ALL PRODUCTS

Notes: Histograms compare treated products and all products (including treated products) using 1997 characteristics. Comparative advantage uses trade shares, concentration is the Herfindahl index of domestic production, and production is measured in logs. Each distribution is normalized separately. The same characteristics are included as year-specific controls in the product-level regressions.

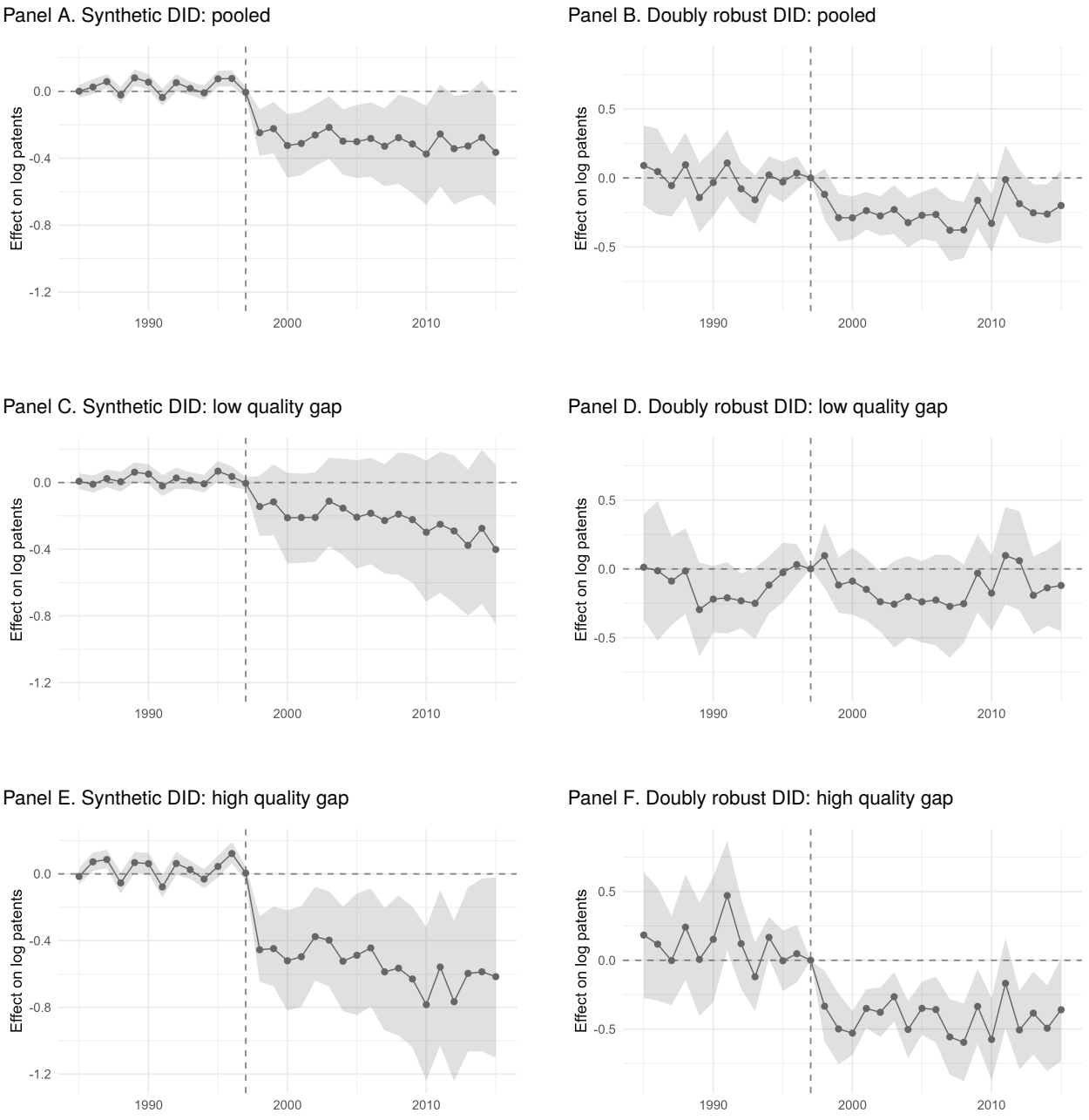


FIGURE A2. PATENT RESPONSES UNDER ALTERNATIVE ESTIMATORS

Notes: The figure estimates corporate patent responses using alternative comparison methods. Both approaches preserve the larger decline in the high-gap group. Synthetic DID pooled treatment effects (standard errors in parentheses) are −0.296 (0.132) overall, −0.227 (0.190) below the quality median, and −0.547 (0.211) above it; the corresponding doubly robust two-period estimates are −0.213 (0.107), −0.067 (0.172), and −0.421 (0.109). Synthetic DID uses a balanced 1985–2015 panel of thirty treated groups and 1,016 donors, residualizing predetermined controls and category shocks, with 999 placebo assignments for 90 percent intervals. Doubly robust event studies compare each year with 1997, conditioning on predetermined characteristics and donor-estimated category–year effects, with analytic 90 percent intervals.

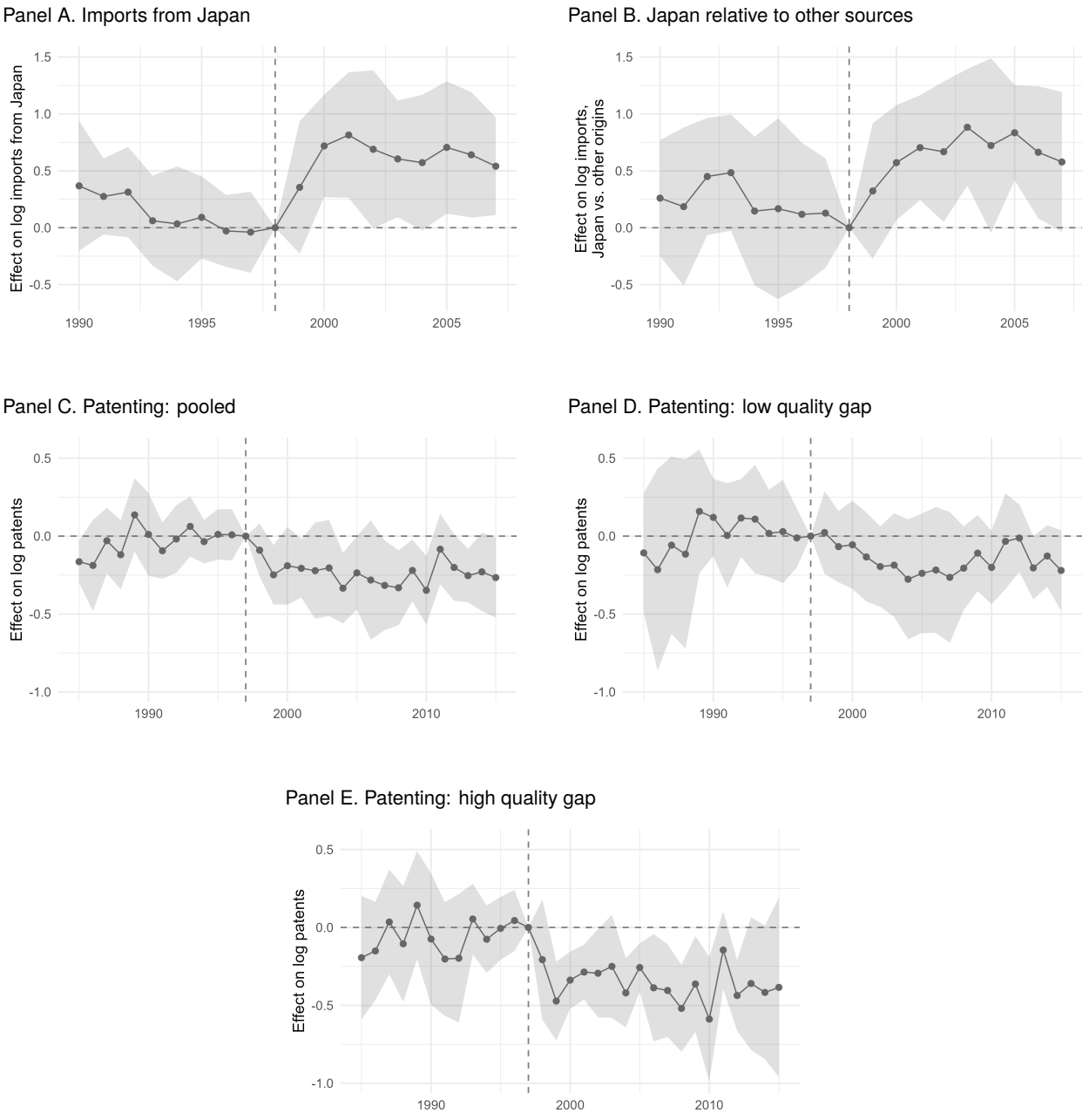


FIGURE A3. RANDOMIZATION-BASED INFERENCE

Notes: Points reproduce the baseline event-study coefficients. Shaded bands are 95 percent pointwise randomization intervals, constructed from studentized statistics across 999 treatment reassignments. Import regressions use 1998 as the reference year; patent regressions use 1997. Fixed effects, controls, samples, and treatment-set sizes follow the corresponding main specifications.

A.3 Financial Conditions and Transport Technologies

Industries with greater debt burdens may have responded more sharply to the Asian financial crisis. I use NICE ValueSearch financial statements to measure liabilities relative to assets in 1997, using 1996 when both accounts are unavailable in 1997. Each patent group's leverage is the average among linked firms weighted by their 1985–1997 contributions to its patent counts. Interacting this measure with year leaves a patent decline of about 16 percent and a larger response in the high-gap group (Figure A4). The innovation response therefore persists after allowing indebted industries to follow different crisis-period paths.

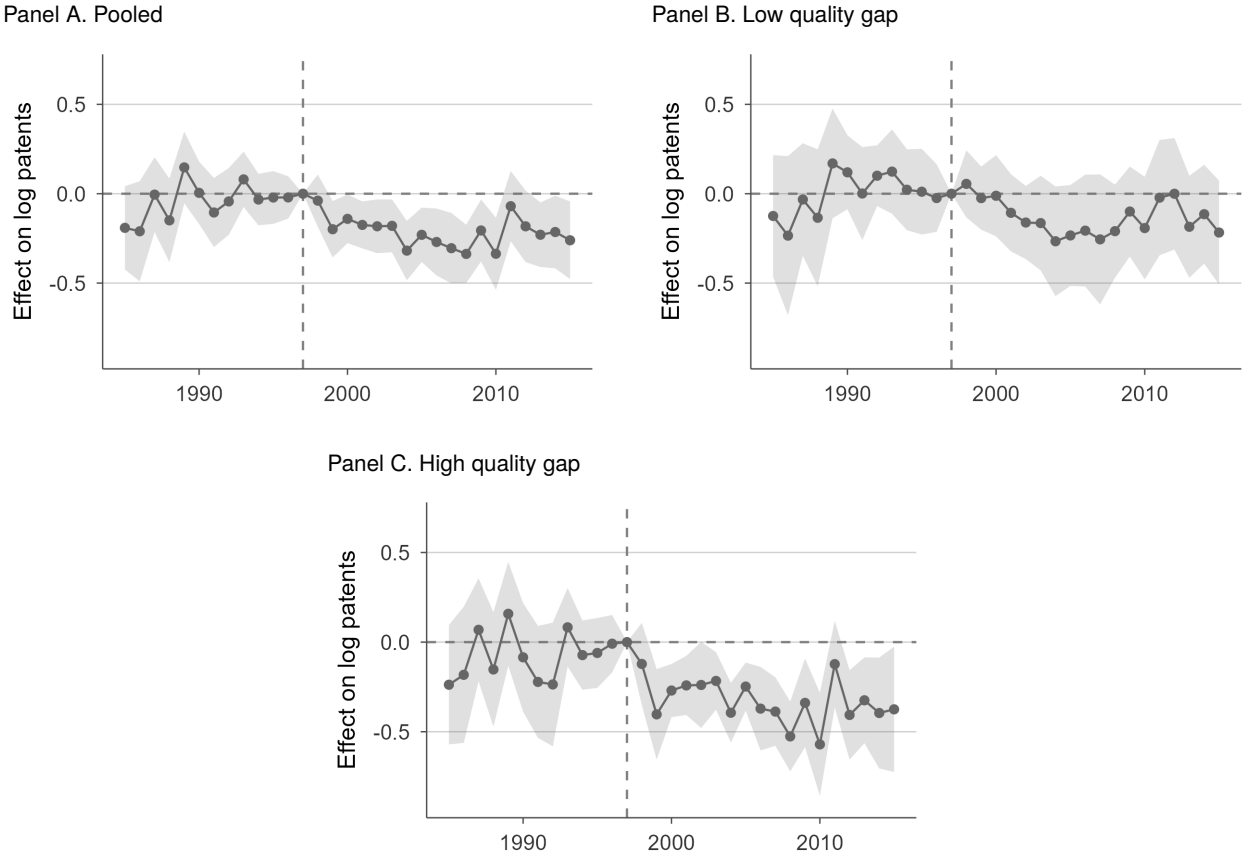


FIGURE A4. PATENT RESPONSES CONTROLLING FOR INITIAL LEVERAGE

Notes: The panels estimate patent responses after allowing annual trends to vary with initial leverage. The decline and its quality ordering persist. Pooled treatment-by-post coefficients (clustered standard errors in parentheses) are −0.174 (0.088) overall, −0.134 (0.166) below the quality median, and −0.258 (0.077) above it. Leverage is liabilities divided by assets from NICE ValueSearch, aggregated using firms' pre-liberalization patent contributions. Samples, fixed effects, and other controls follow Figures 4 and 5. Bands are 90 percent concordance-clustered confidence intervals relative to 1997.

Vehicle technologies represent a prominent part of the liberalized products and could disproportionately influence the aggregate result. Excluding class B60 leaves a patent decline of about 16 percent (Figure A5), showing that the retreat extends to other exposed technologies.

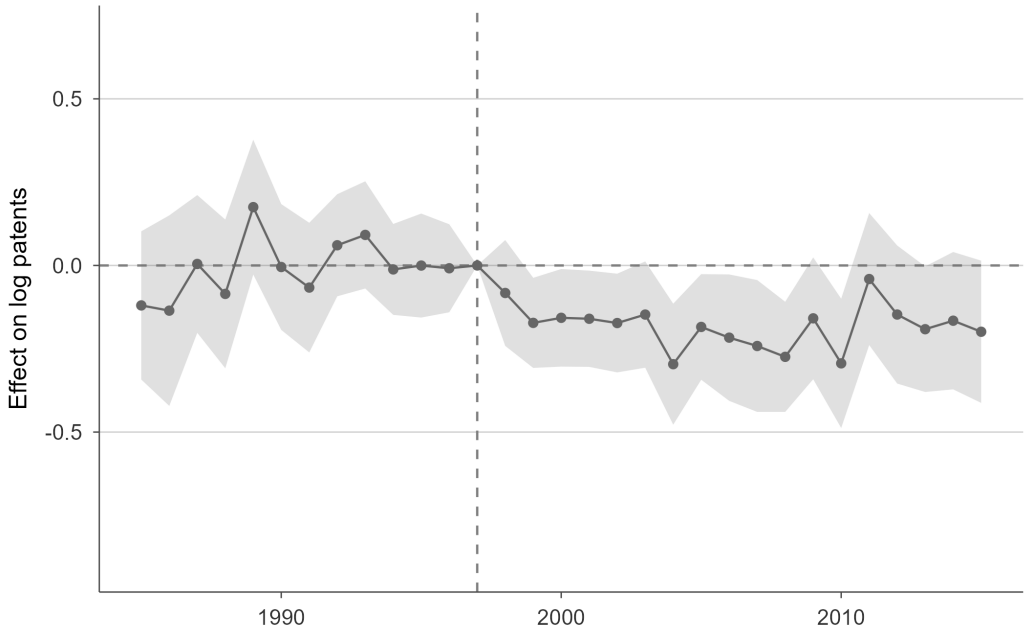


FIGURE A5. PATENT RESPONSES EXCLUDING VEHICLE TECHNOLOGIES

Notes: The figure repeats the patent event study excluding vehicle technologies in class B60. Patenting continues to decline outside this exposed category. The pooled treatment-by-post coefficient is -0.176 (0.089). The sample contains 106,708 observations in 3,599 patent groups. Fixed effects and controls follow Figure 4. Bands are 90 percent concordance-clustered confidence intervals relative to 1997.

A.4 Non-Corporate Patenting as a Placebo

Common changes in technological opportunities could reduce patenting by applicants with different exposure to product-market competition. I therefore apply the corporate specifications to public bodies, research institutes, and individuals, using the same patent-activity criteria separately for these applicants.

Non-corporate patenting shows a smaller decline and a weaker relationship with the initial quality gap (Figure A6). These differences support a role for commercial competition in the corporate response.

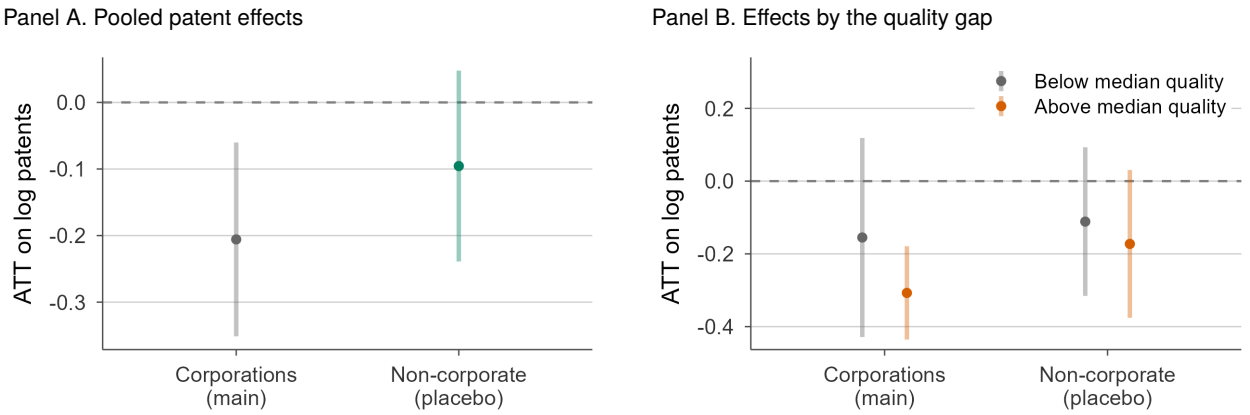


FIGURE A6. NON-CORPORATE PATENTING AS A PLACEBO FOR COMMON TECHNOLOGICAL TRENDS

Notes: Non-corporate applicants show a smaller patent decline and a weaker quality gradient. Pooled treatment-by-post coefficients (clustered standard errors in parentheses) are −0.206 (0.088) for corporations and −0.096 (0.087) for non-corporate applicants; the latter group's quality-split estimates are −0.111 (0.124) below and −0.173 (0.123) above the median. Panel (A) applies the pooled specification of Figure 4 to corporate and non-corporate filings. Non-corporate applicants are public bodies, research institutes, and individuals. The corporate sample has 107,022 observations and 3,612 patent groups (36 treated); the non-corporate sample has 111,256 observations and 3,678 groups (37 treated). Panel (B) follows Figure 5, with a cutoff of 1.238 and jointly estimated effects for the two quality groups. The corporate sample has 17 below-median and 16 above-median treated groups; the non-corporate sample has 17 in each group. The corporate sample has 107,023 observations. Each sample requires five positive-count years before and after the announcement. The post period is 1998–2015. Bars are 90 percent confidence intervals clustered by concordance cluster.

A.5 Differences in Patent Importance

Equal patent counts may hide changes in the importance of individual inventions. Following the text-similarity approach of Kelly et al. (2021), I weight Korean patents by their similarity to subsequent inventions relative to preceding inventions. The measure compares a ten-year forward window with a five-year backward window. Figure A7 reports results using log-transformed importance and its within-year percentile rank. The decline therefore also appears when more influential inventions receive greater weight.

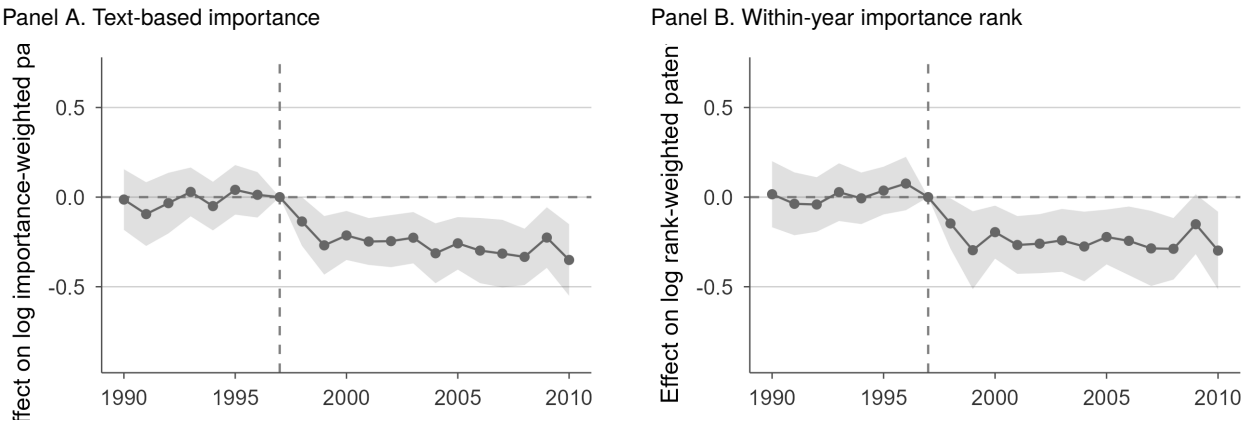


FIGURE A7. PATENT RESPONSES WEIGHTED BY INVENTION IMPORTANCE

Notes: The panels estimate patent responses using text-based invention importance as a weight. The decline persists when more influential inventions receive greater weight. Pooled treatment-by-post coefficients (clustered standard errors in parentheses) are -0.250 (0.076) and -0.253 (0.080), respectively. The outcome is the log of importance-weighted patent counts. The first panel uses log-transformed importance, winsorized at the 0.5th and 99.5th percentiles within filing year; the second uses its within-year percentile rank. The estimation period is 1990–2010, with at least five positive years on each side of 1997. The samples contain 71,580 and 71,645 observations, respectively. Fixed effects and controls follow the baseline patent model. Bands are 90 percent concordance-clustered confidence intervals.

B Product Quality: Measurement and Heterogeneity

B.1 Relative Prices and Inferred Quality

Unit values reflect both quality and prices. Equation (4) uses relative sales to infer the quality consistent with observed unit values, following Khandelwal et al. (2013). I apply it to the 1996–1997 US import data of Schott (2008), using the product coverage and weights defined in Section 4.4.

Figure B1 compares relative unit values and quality across 2,498 HS6 products, including 39 treated products. Median log unit-value ratios are 0.535 for all products and 0.443 for treated products; median quality gaps are 0.854 and 1.180. Japanese unit values exceed Korean values in 29 treated products, while inferred Japanese quality is higher in 35. Sales information therefore strengthens the measured Japanese advantage. Treated products span a wide range, from near or below parity to a substantial Japanese lead.

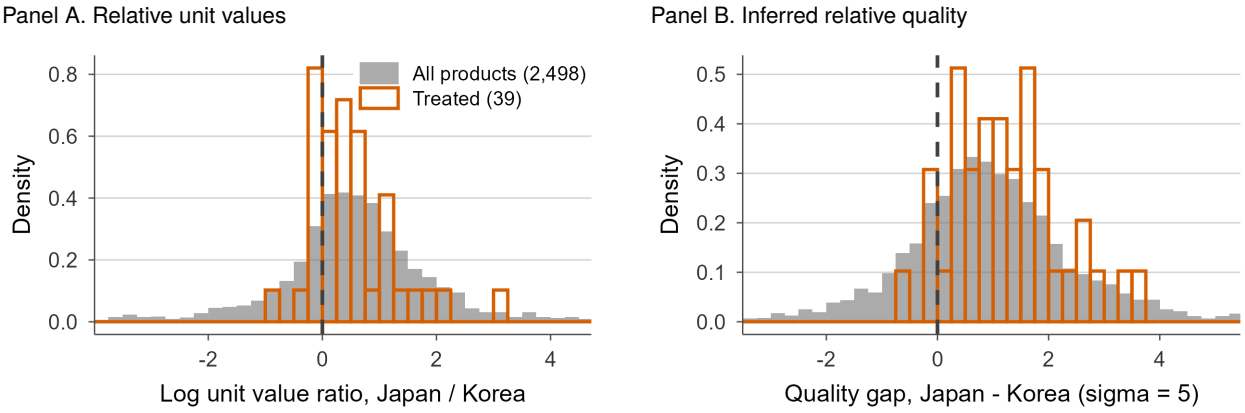


FIGURE B1. PRE-LIBERALIZATION RELATIVE PRICES AND QUALITY IN US IMPORTS

Notes: Gray filled histograms show all 2,498 products; orange outlines show the 39 treated products with observed values. Each distribution is normalized to a density, with bin width 0.25. Values pool 1996–1997 US imports from Japan and Korea within common HTS10 lines and aggregate using combined import-value weights. Quality uses $\sigma = 5$. The displayed horizontal range is the first through 99th percentile of the full distribution; observations outside that range remain in the density calculation. The vertical dashed line marks parity.

B.2 Relationship with Independent Indicators

The quality gap is positively related to the patent-based gap, the Japan–Korea RCA difference, and import penetration (Figure B2). These modest associations support interpreting it as competitive standing while showing that sales, patenting, and export specialization capture different dimensions. Import penetration also reflects domestic demand, production, and the restrictions themselves.

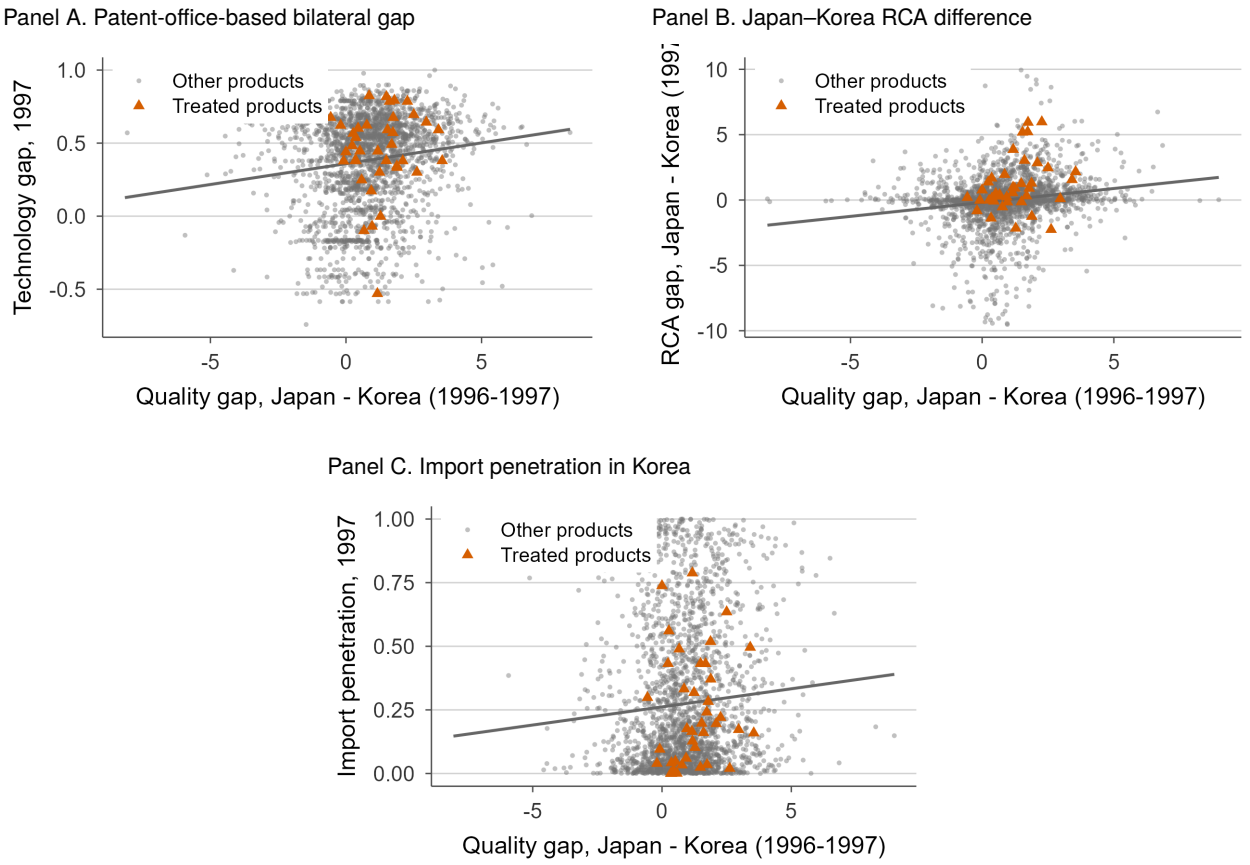


FIGURE B2. PRODUCT QUALITY AND INDEPENDENT INDICATORS OF BILATERAL STANDING

Notes: Each point is an HS6 product; orange triangles denote treated products. Lines are full-sample OLS fits. Panel (A) uses the patent-based gap in Appendix B.3 (2,266 products, 38 treated). Panel (B) uses the 1997 RCA difference (2,473 products, 39 treated), excluding 24 products with absolute differences above ten. Panel (C) uses 1997 world imports divided by imports plus domestic production in a common currency (2,256 products, 39 treated). Full-sample Pearson correlations are 0.136, 0.172, and 0.075 (all $p < 0.001$); rank correlations are 0.161, 0.233, and 0.098. Treated-product Pearson correlations are 0.107 ($p = 0.52$), 0.302 ($p = 0.061$), and 0.112 ($p = 0.50$).

B.3 An Alternative Patent-Based Bilateral Gap

Patent activity provides an alternative measure of relative technological standing that is independent of US prices and sales. Using Google Patents BigQuery, I count documents published at the Japanese and Korean patent offices during 1970–1997, sum their CPC symbols, aggregate to IPC subclasses, and apply the Lybbert and Zolas (2014) product weights. For patent group g, the gap is

$$\text{TechGap}_g = \frac{P_g^{\text{JPO}} - P_g^{\text{KIPO}}}{P_g^{\text{JPO}} + P_g^{\text{KIPO}}}. \tag{B.1}$$

It ranges from -1 to $+1$, with larger values indicating greater activity at the Japanese office. The counts include applicants of all nationalities and reflect filing coverage as well as technological standing.

Patenting falls by about 6 percent below the median patent-based gap and 30 percent above it (Figure B3). The larger retreat where Japan held a greater advantage therefore also appears with a measure based on patent activity.

B.4 Import Responses by the Initial Quality Gap

A larger initial Japanese advantage could amplify the competitive shock as access to the Korean market expands. I examine this relationship by estimating import responses across groups defined by their initial quality gap, retaining the comparison and controls of Figure 2(A).

Japanese imports increase more where Japan initially held a larger quality lead (Figure B4). The pooled estimates imply increases of about 40 percent below the median and 136 percent above it. Point estimates also rise monotonically across quality terciles, linking the intensity of import competition to Japan's initial competitive advantage.

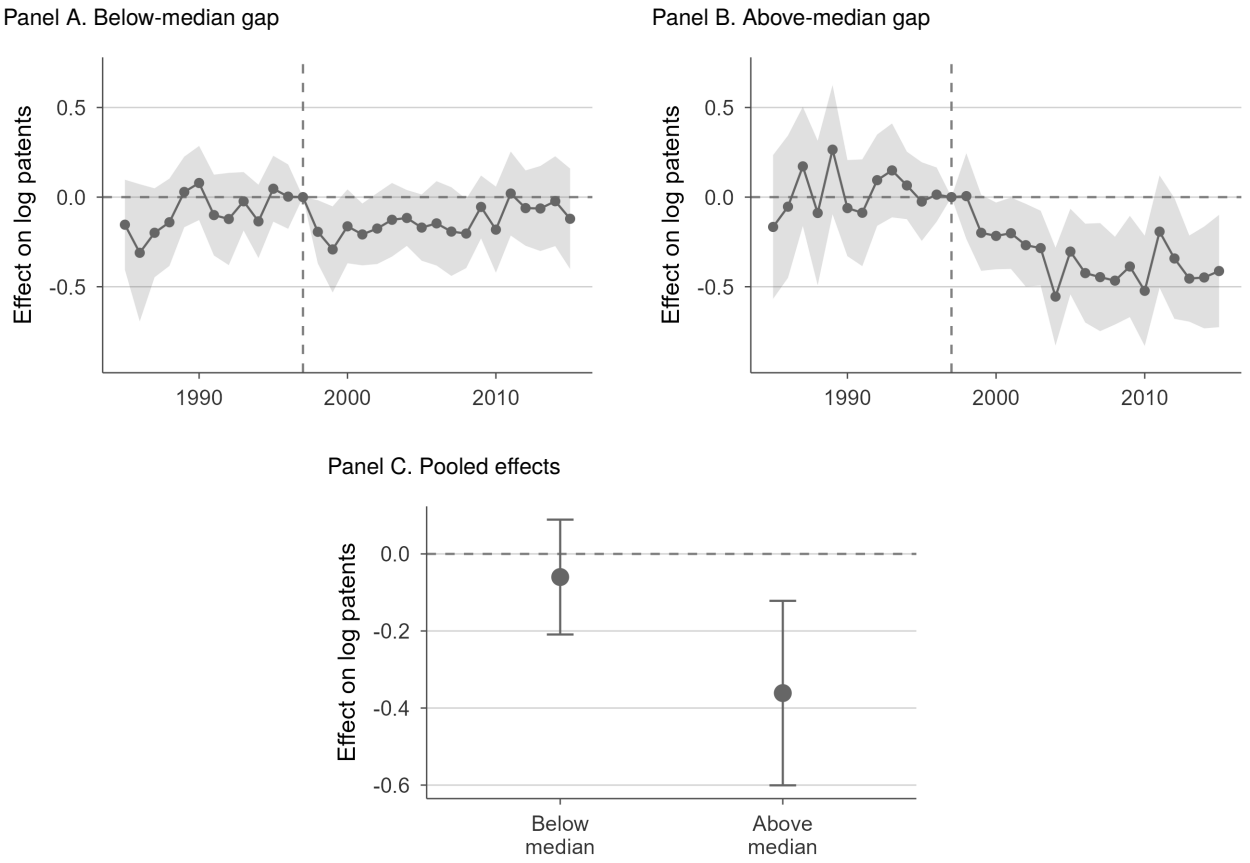


FIGURE B3. PATENT RESPONSES BY THE PATENT-BASED JAPAN–KOREA GAP

Notes: The figure estimates patent responses by Japan's initial patent-based technological advantage. Patenting falls more in the above-median group. Pooled treatment-by-post coefficients (clustered standard errors in parentheses) are −0.060 (0.091) below and −0.361 (0.146) above the median. The cutoff of 0.486 divides 38 treated patent groups into eighteen eligible groups on each side. The sample contains 107,023 observations, 3,612 patent groups, and 3,492 concordance clusters. Outcomes, fixed effects, and controls follow Figure 5; 1997 is the reference year and 1998–2015 the post period. Intervals are 90 percent confidence intervals.

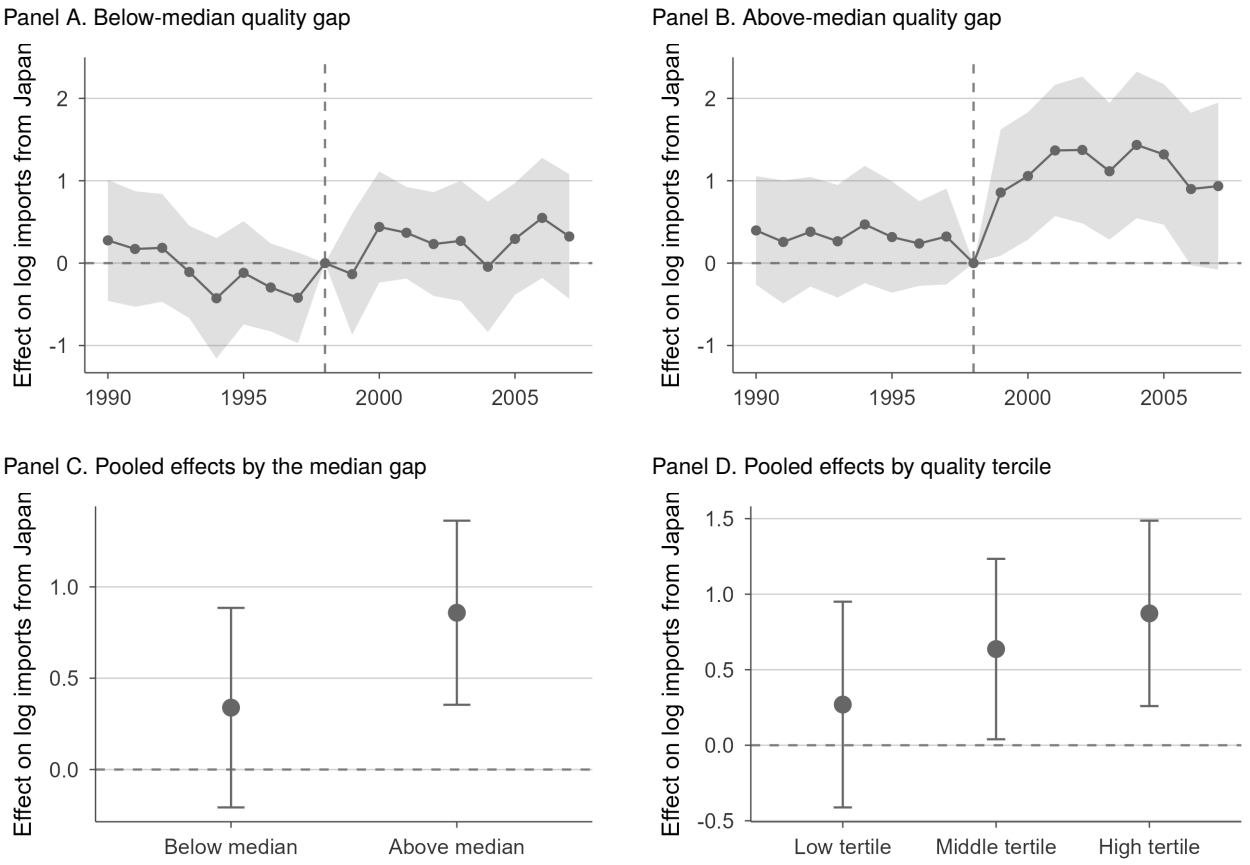


FIGURE B4. JAPANESE IMPORT RESPONSES BY THE BILATERAL QUALITY GAP

Notes: Panels (A)–(B) estimate annual import responses by the initial quality gap; panels (C)–(D) report single treatment-by-post coefficients for the median and tercile splits. Import-response point estimates increase with Japan's initial quality advantage. The median-split coefficients (clustered standard errors in parentheses) are 0.339 (0.332) and 0.858 (0.306); tercile estimates are 0.269 (0.414), 0.637 (0.363), and 0.873 (0.373). The median cutoff is 1.176 and tercile cutoffs are 0.698 and 1.659. The sample contains 54,628 observations and 3,142 trade-group clusters, including 35 treated groups with measured quality. Fixed effects and controls follow Figure 2(A). Intervals are 90 percent confidence intervals; the reference year is 1998 and the post period is 1999–2007.

C Patent Data Construction and Technology Classification

C.1 Harmonizing Corporate Applicant Codes

Corporate restructuring can interrupt an applicant's recorded patent history even when its inventive activity continues under a successor. To follow firms through these changes, I consolidate codes belonging to the same legal entity using corporate or business registration numbers and verified name histories, and trace mergers, acquisitions, and spin-offs by cross-checking KIPRIS reassignment records against corporate histories and public filings.

The domestic corporate sample excludes linked histories that became wholly foreign-owned during the sample period, including confirmed business successors. This restriction is applied to the entire history so that changes in corporate ownership do not alter the composition of domestic incumbents over time. Table C1 records the successive stages of applicant harmonization, technology construction, and firm-sample selection.

C.2 Construction of CPC Cells and Treatment Definitions

Direct product–technology links provide a check on the probabilistic concordance. The CPC hierarchy runs from sections (A) through classes (A47), subclasses (A47J), and main groups (A47J27) to subgroups (A47J27/086), permitting selective aggregation of sparse technologies.

An initial unit remains separate if it has at least three Korean corporate incumbents in 1985–1992 and positive patent counts in at least fifteen years during 1993–2010. Units with no pre-period incumbent are excluded. Remaining sparse units are pooled within main groups, then subclasses, under the same activity criteria. The pooled incumbent requirement sums the incumbent counts of its constituent initial units. Table C1 summarizes the resulting technology coverage.

Three nested treatment definitions vary the breadth of the product–technology link. The conservative set uses product-specific assessments and reviewed direct links. The standard set adds supporting KIPO product–technology links, and the inclusive set adds broader links. Their sizes appear in Table C1; the firm analysis uses the standard set. Cells outside each set serve as controls.

TABLE C1—CONSTRUCTION OF APPLICANT HISTORIES, TECHNOLOGY CELLS, AND FIRM SAMPLES

Construction stage	Count
Panel A. Corporate applicant histories	
Applicant codes	97,625
Legal entities after identity linkage	97,364
Firm histories after succession linkage	97,240
Histories excluded for full foreign ownership	12
Domestic firm histories	97,228
Applications in the domestic corporate sample	1,885,706
Panel B. Technology classification	
Initial technology units	7,870
Initial units represented in retained cells	6,836
Final CPC cells	5,617
Treated cells: conservative / standard / inclusive	132 / 206 / 271
Panel C. Firm estimation samples	
Exposed candidates for matching	296
Exposed firms receiving a match	276
Treated–control pairs	811
Distinct matched controls	537
Between-firm estimation: exposed / control firms	238 / 459
Within-firm estimation: high-share / low-share firms	253 / 251

Notes: Panel (A) follows corporate identifiers through legal-entity and documented succession linkage and the domestic-ownership restriction. Panel (B) applies the activity criteria in Appendix C.2; treatment sets are nested. Panel (C) applies the sample and matching rules in Appendix E.1. Matched firms and estimation firms are reported separately because the PPML comparison uses patenting variation within the fixed-effect groups. Controls can serve in more than one matched set. The within-firm sample contains 504 firms, divided at the common other-field-share threshold.

C.3 Patent Responses and Quality Heterogeneity

I estimate log corporate patent counts with cell, IPC-class-by-year, and size-bin-by-year fixed effects, plus year interactions with log pre-period incumbent counts. Size bins split total 1985–2015 patent counts at 1,000. Cells require five positive-count years on each side of 1997; logs use positive observations.

Patenting declines under all three definitions (Figure C1). Estimated reductions range from about 31 percent under the inclusive definition to 49 percent under the conservative definition. The stronger response for the most direct links is consistent with closer technological exposure to the liberalized products.

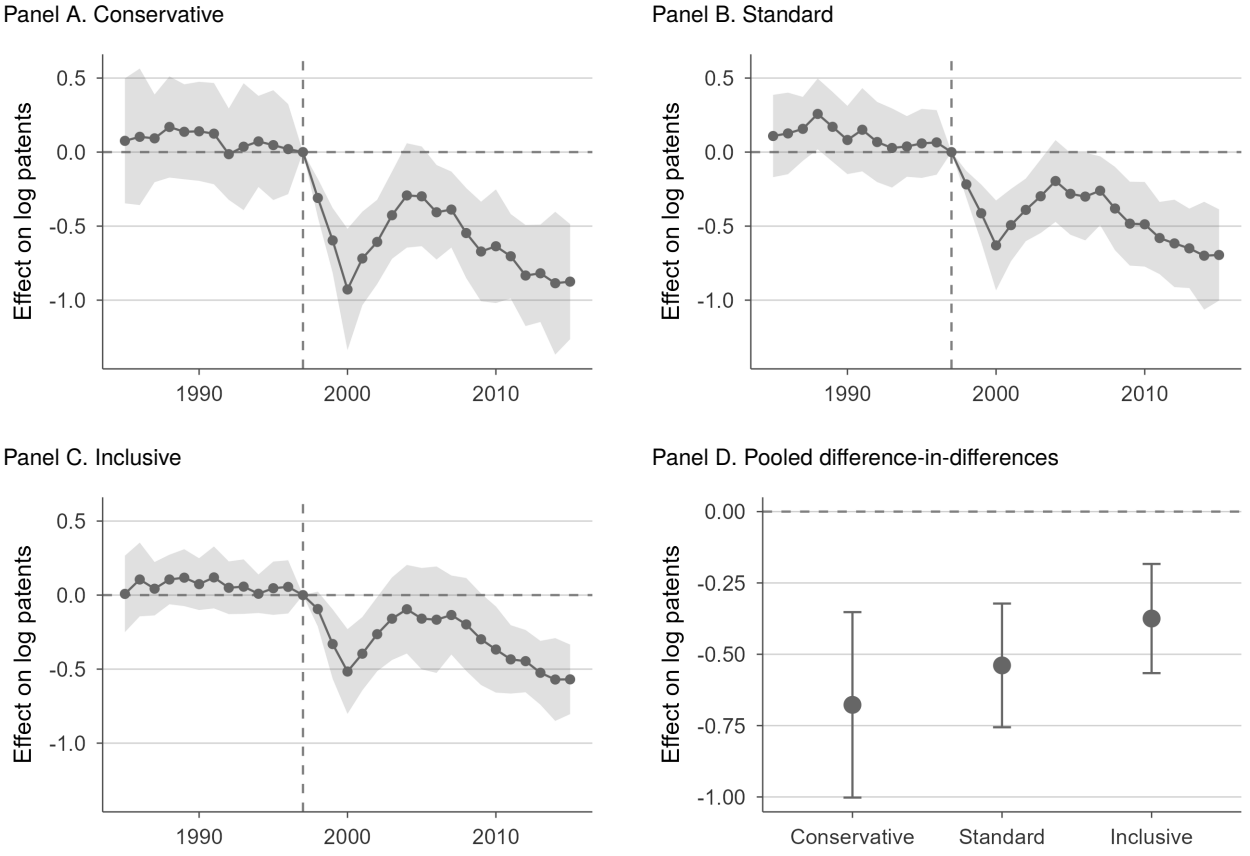


FIGURE C1. PATENT RESPONSES UNDER ALTERNATIVE CPC TREATMENT DEFINITIONS

Notes: The figure estimates patent responses under three nested definitions of technological exposure. Declines are greater for the most direct product links. Pooled treatment-by-post coefficients (clustered standard errors in parentheses) are −0.677 (0.198), −0.539 (0.132), and −0.375 (0.116) for the conservative, standard, and inclusive sets. All models use 158,951 observations, 5,607 cells, and 112 IPC-class clusters. Panels (A)–(C) plot annual coefficients relative to 1997; panel (D) reports the single treated-by-post coefficient from each corresponding pooled regression. The sample covers 1985–2015 and the post period is 1998–2015. All treated cells satisfy the positive-count eligibility rule. Fixed effects and controls are described above. Intervals are 90 percent confidence intervals with IPC-class-clustered standard errors.

I then examine whether these responses vary with Japan's initial quality advantage. Each treated cell receives the average gap of its linked treated products with observed quality, and I split cells at the median within each treated set. Patenting falls more above the median under all three definitions (Figure C2). The ordering holds as the definition expands from direct product links to broader technological exposure.

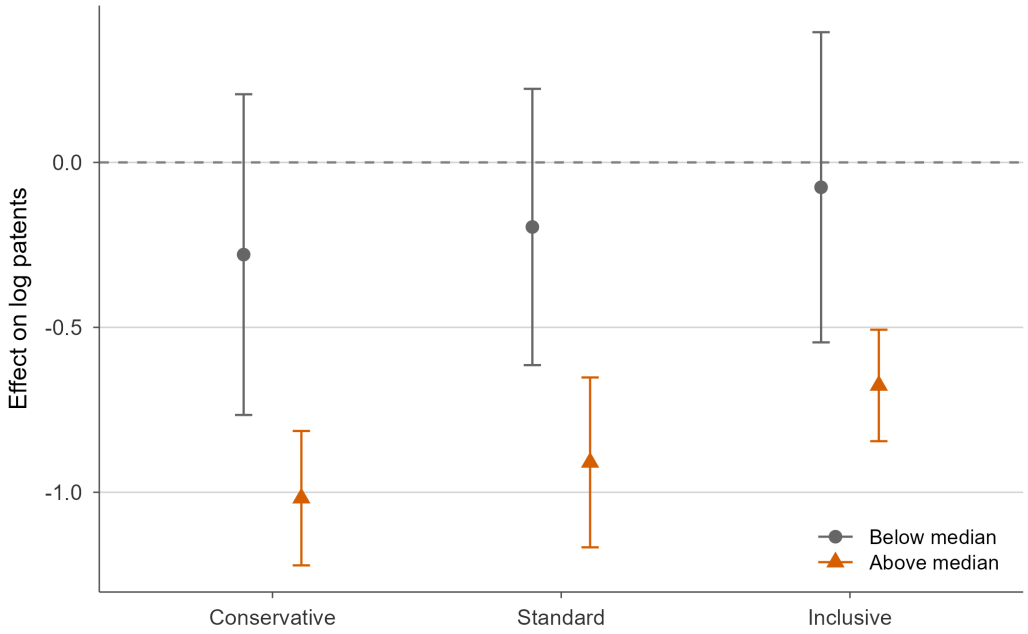


FIGURE C2. CPC PATENT EFFECTS BY RELATIVE QUALITY

Notes: Effects above and below the median quality gap are estimated jointly within each treatment definition. Patent declines are larger in high-gap technologies. The differences between the above- and below-median coefficients (clustered standard errors in parentheses) are -0.738 (0.340), -0.714 (0.362), and -0.601 (0.347), respectively. Median gaps are 1.730 (conservative), 1.695 (standard), and 1.158 (inclusive). The respective samples contain 63 and 69, 100 and 102, and 130 and 137 cells with measured quality below and above the median. Each specification uses the same 158,951-observation CPC sample and controls as Figure C1. Bars are 90 percent confidence intervals.

D Product Composition and Domestic Production

D.1 Multi-Product Establishments

Multi-product establishments can adjust across products when one activity faces greater competition. Figure D1 reports their production and employment responses, adding the final-goods coefficient to its interaction with multi-product status in the baseline regression. Relative to single-product establishments, the production decline is smaller, while the employment decline is slightly larger.

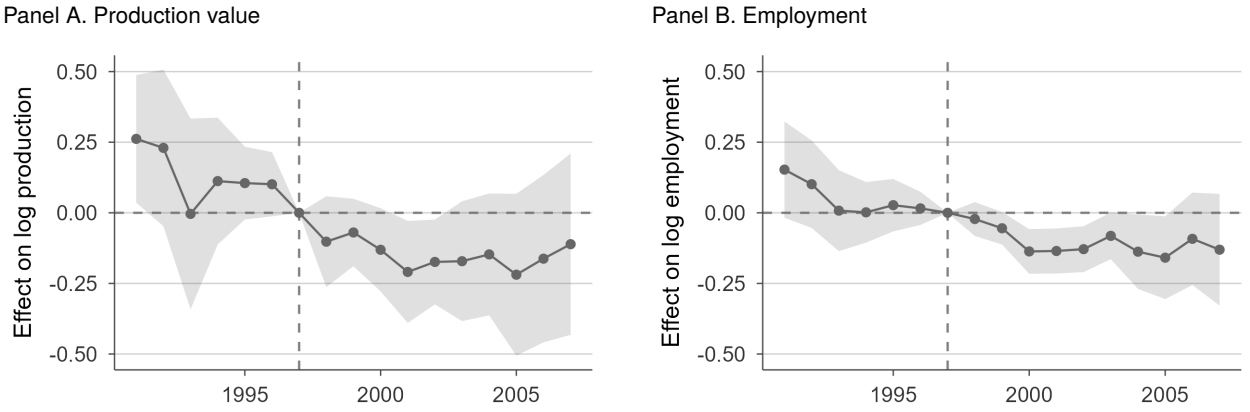


FIGURE D1. DOMESTIC PRODUCTION AND EMPLOYMENT: MULTI-PRODUCT ESTABLISHMENTS

Notes: The figure estimates production and employment responses among multi-product establishments. Production declines less than among single-product establishments, while employment losses are comparable. Mean annual post-period coefficients are -0.150 for production and -0.108 for employment. Sample, treatment, fixed effects, and controls follow Figure 3. Multi-product status indicates at least two products in 1997 or 1998. Each coefficient sums the final-goods effect and its multi-product interaction.

D.2 Japanese Subsidiaries and Product Composition

Japanese-owned manufacturers in Korea were concentrated in intermediate-goods categories linked to the ISDP. Their presence complicates the interpretation of domestic production in those categories because the Mining and Manufacturing Survey cannot distinguish them from Korean-owned establishments. This motivates the focus on final goods in Section 4.2.

I obtained the available editions of Tōyō Keizai's *Kaigai Shinshutsu Kigyō Sōran* (*Overseas Enterprise Directory*) and digitized all entries for subsidiaries in Korea. These records show that Japanese subsidiaries were more prevalent in intermediate-goods manufacturing both before and after abolition (Figure D2). This composition supports examining final-goods establishments to trace the response of domestic producers.

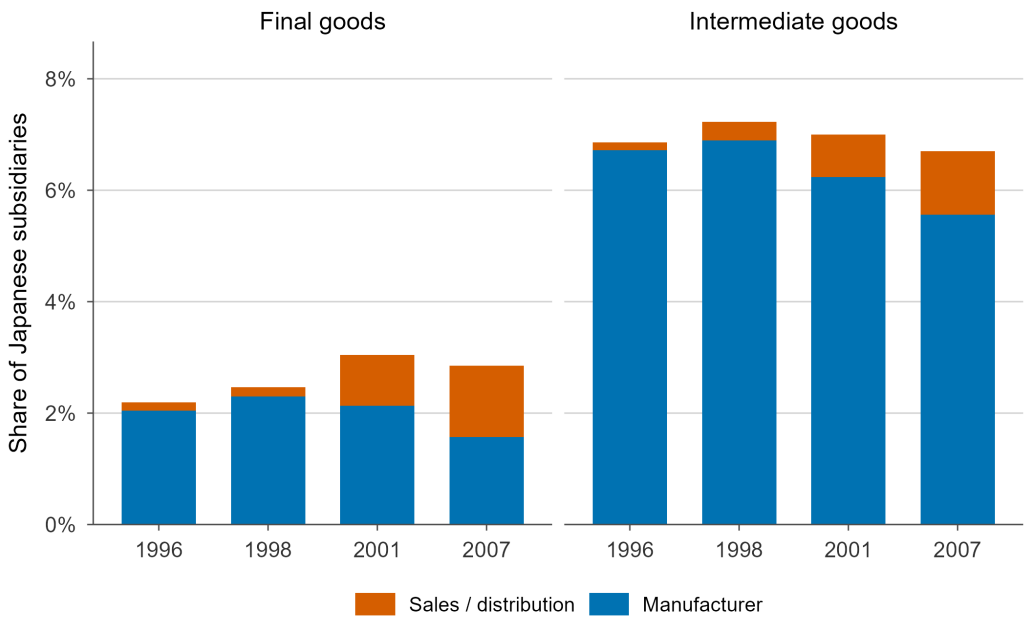


FIGURE D2. PRODUCT COMPOSITION OF JAPANESE SUBSIDIARIES IN KOREA

Notes: Shares of all Japanese-owned subsidiaries in each directory edition, separating final- and intermediate-goods links to the ISDP. Orange denotes sales and distribution; blue denotes manufacturing. Denominators are 685, 609, 657, and 701 subsidiaries in 1996, 1998, 2001, and 2007. Presence runs from the earlier of establishment or first listing through the last listing. Product category is fixed across editions; a subsidiary is classified as a manufacturer if at least half its observed listings report manufacturing. Source: Tōyō Keizai directories, covering 1,116 subsidiaries and 2,652 subsidiary-edition observations.

E Firm Portfolios, Estimation, and Reallocation

E.1 Samples and Matching

Pre-period patenting is the sum of patent counts across classified technologies during 1985–1997. Exposure is the share of these counts in the 206 treated cells. Total annual applications count unique filings, including applications whose classification symbols do not enter the retained-cell portfolio.

For the between-firm comparison, candidates have exposure of at least 5 percent and pre-period patent counts of at least five; possible controls have exposure at most 1 percent and the same patent-count requirement. A firm's main field is the WIPO field with the largest pre-period patent count. I match firms whose main fields belong to the same sector and require an exact match on whether they filed in each of four windows: 1986–1988, 1989–1991, 1992–1994, and 1995–1997. A control's pre-period patent count must be between one-third and three times the exposed firm's count. Within this pool, I select firms with similar patenting volume and technological knowledge bases, using the mean rank of Mahalanobis distance in log patent counts and cosine distance between IPC-class portfolios. Each exposed firm receives up to three controls, with replacement across sets.

Table C1 reports the number of candidates, matched firms, and firms contributing to estimation. Weights adjust for initial patenting and reuse across matched sets. Figure E1 shows how matching aligns pre-treatment characteristics. Each control follows the share group of its matched exposed firm in the heterogeneity analysis.

The within-firm analysis includes firms with exposure of at least 1 percent that filed at least one patent containing treated cells and at least one containing none during 1985–1997. Firms must have at least two years of patenting in 1993–2007 and an observed other-field share. The common other-field-share cutoff is 70.79 percent. The exposure reference of 10.75 percent is held fixed across the firm analyses.

I also repeat matching within firms linked to 1997 ValueSearch financial statements and allow annual paths to vary with initial log employment, log revenue, leverage, and ROA. Figure E2 shows that the heterogeneity in total patenting changes little after controlling for these financial characteristics: high-share firms sustain patenting, while low-share firms reduce it.

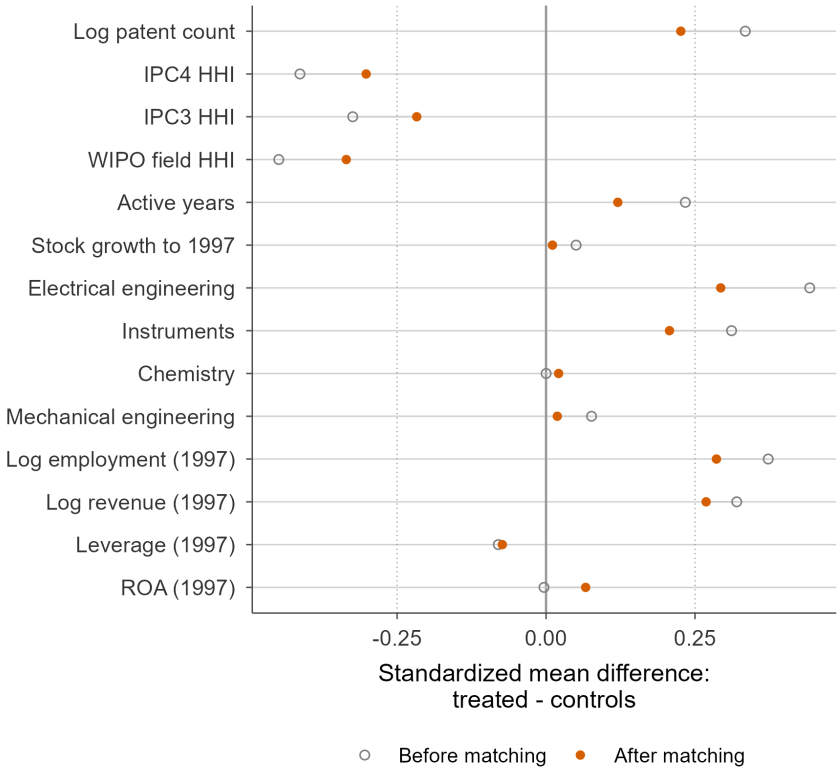


FIGURE E1. PRE-TREATMENT BALANCE IN THE MATCHED-FIRM SAMPLE

Notes: The figure compares pre-treatment characteristics of exposed and low-exposure firms before and after matching. Points show standardized differences in means, divided by the square root of the mean variance in exposed firms and the eligible control pool. Each firm is counted once. Sector indicators denote any pre-period patenting in that sector. The final four rows use observed 1997 ValueSearch data within these groups, with missing values excluded separately for each characteristic. Leverage is total liabilities divided by assets; ROA is operating income divided by assets. Financial characteristics do not enter matching. The sectors of firms' main fields and filing activity in the four pre-period windows are matched exactly; the remaining selection uses pre-period patent counts and portfolio similarity.

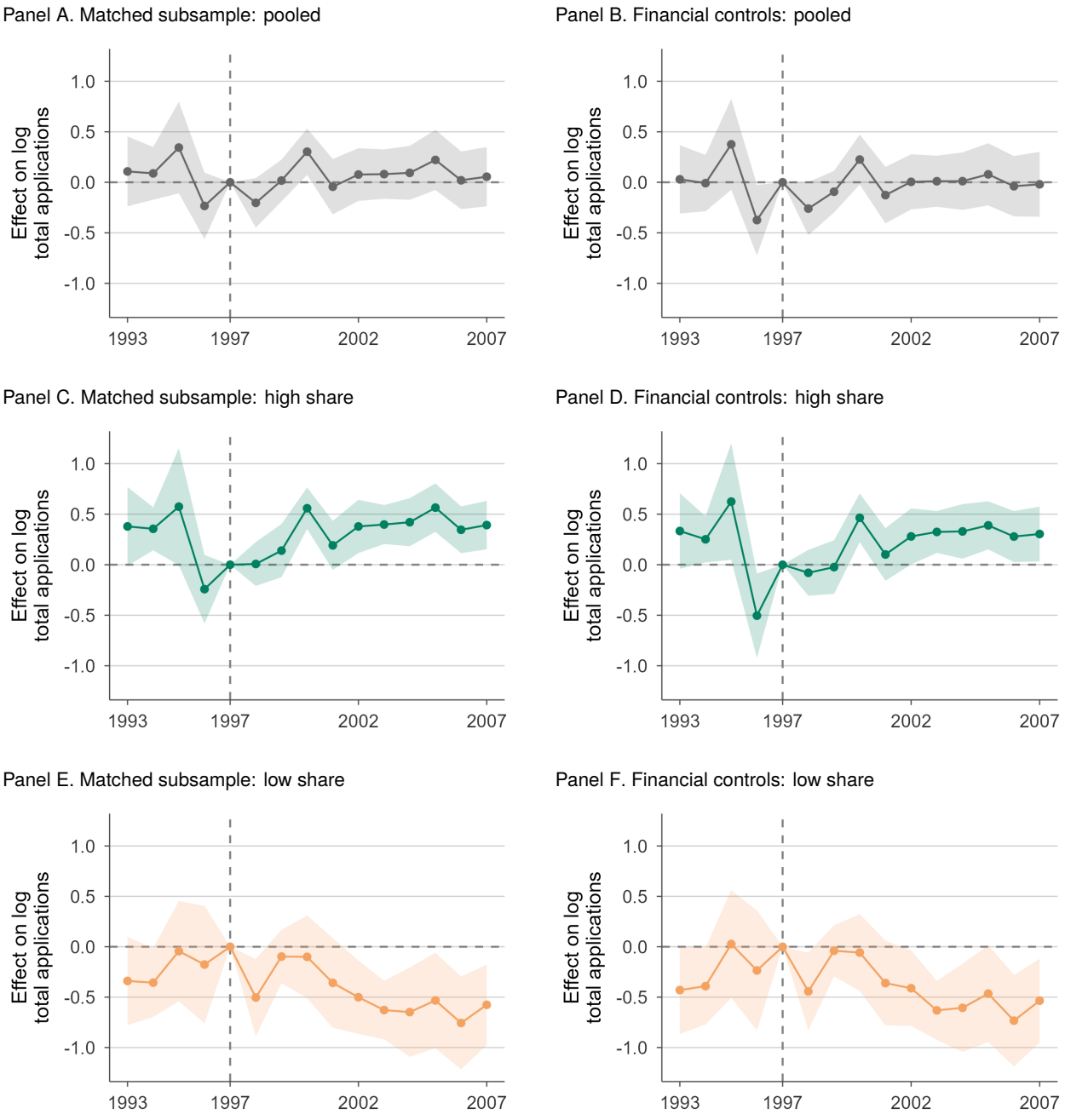


FIGURE E2. TOTAL PATENTING AND INITIAL FINANCIAL CHARACTERISTICS

Notes: Left panels repeat the main patent-based matching procedure among firms with 1997 ValueSearch financial data. Right panels add year interactions with 1997 log employment, log revenue, leverage, and ROA, with common slopes across share groups. Leverage is total liabilities divided by assets; ROA is operating income divided by assets. Left panels use 96 exposed firms (50 high-share, 46 low-share), 168 controls, and 4,935 observations. Complete financial cases in the right panels give 95 exposed firms (49 high-share, 46 low-share), 166 controls, and 4,872 observations. On this common complete-case sample, adding financial controls changes high- and low-share post-period means from 0.287 and −0.471 to 0.237 and −0.428. Other settings follow Figure 6(A), including inverse initial-patent-count and reuse weights, the 70.79 percent share cutoff, and the 10.75 percent reference exposure. Bands are 90 percent firm-clustered confidence intervals relative to 1997.

E.2 Raw Changes in Patenting

Changes in observed patenting provide a direct view of the two adjustment margins. Figure E3 compares mean annual patenting before and after liberalization: the distribution of total growth among exposed firms overlaps substantially with the overall distribution, while growth in their treated technologies is generally lower than growth in untreated technologies. This pattern connects the regression evidence to the underlying changes in firms' portfolios.

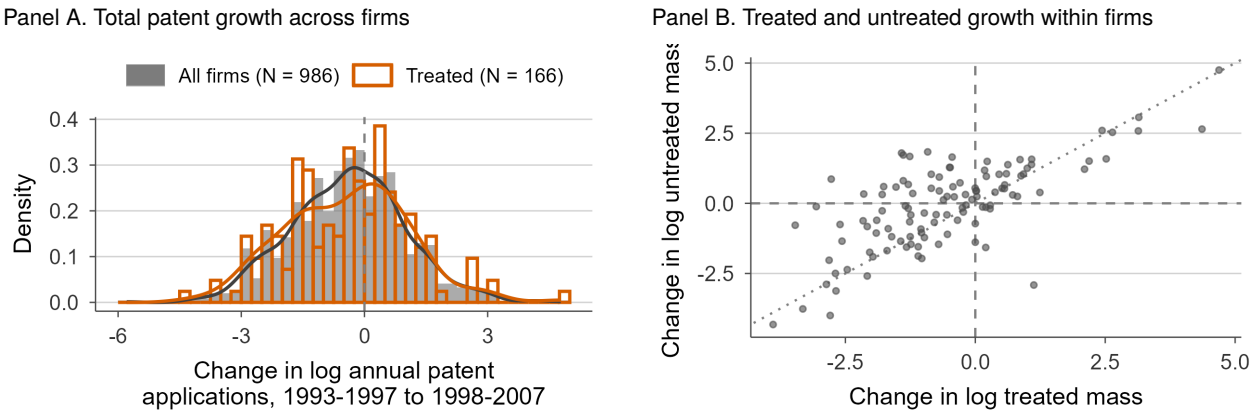


FIGURE E3. RAW CHANGES IN ANNUAL PATENTING

Notes: The figure compares changes in total patenting and in the treated and untreated portions of exposed firms' portfolios. Growth is the change in log mean annual patenting from 1993–1997 to 1998–2007. Panel (A) counts unique applications; panel (B) uses patent counts in treated and untreated cells. Panel (A) requires at least five applications in 1993–1997 and one afterward, with exposure of at least 5 percent for the exposed subset. Panel (B) covers 113 exposed firms with positive patenting in both categories in each period; mean log growth is -0.545 in treated and -0.057 in untreated technologies.

E.3 Weighting by Initial Patenting and the PPML Estimand

Initial patenting is highly concentrated: the two largest exposed candidates account for about 40 percent of pre-liberalization patent counts (Figure E4). In unweighted PPML, such large portfolios can dominate the estimated response. Weighting by inverse pre-period patenting makes each firm's contribution proportional to its change relative to its initial scale. The following benchmark shows how this weighting relates to the average multiplicative response.

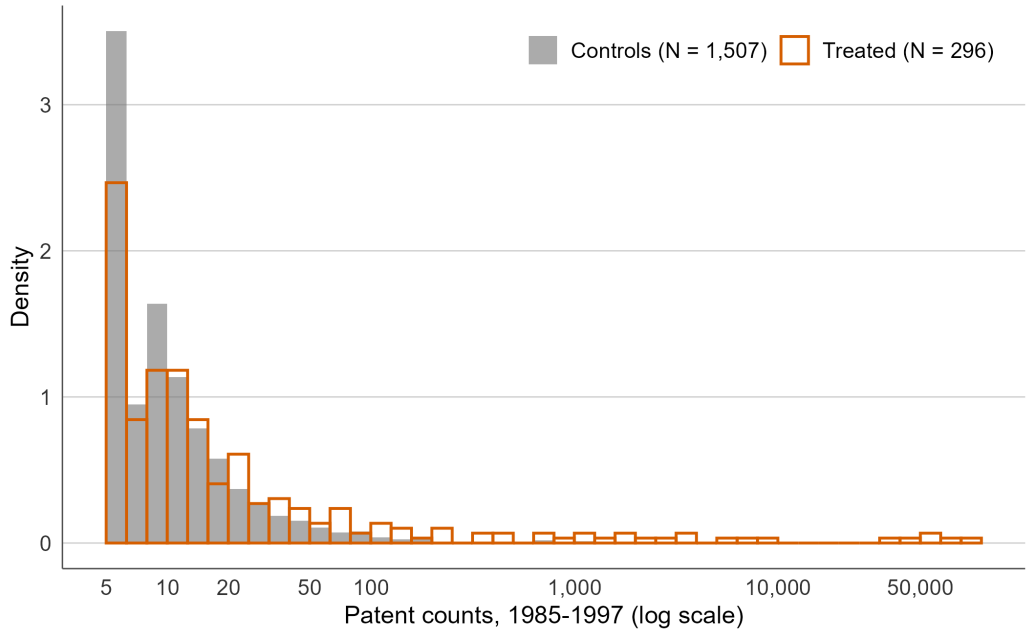


FIGURE E4. DISTRIBUTION OF PRE-LIBERALIZATION PATENTING

Notes: The figure compares cumulative patent counts over 1985–1997 for 296 exposed candidates with exposure of at least 5 percent and 1,507 eligible controls with exposure at most 1 percent, before matching. Both groups have pre-period patent counts of at least five and belong to the pre-period filing cohort. Histograms use common bins of width 0.1 in log-base-ten patent counts and are normalized separately to unit area. The horizontal axis is logarithmic. Among exposed candidates, the median patent count is 11 and the largest is 92,390; the largest two account for 39.8 percent of candidate patenting and 26.6 percent after matching.

PPML estimates the conditional mean in levels, accommodates zero outcomes, and allows the mean to respond proportionally to treatment. Let M_i be firm i's pre-liberalization patent count. With weights $w_i = 1/M_i$, the score for a regressor x_{it} is

$$\sum_{i,t} x_{it} \left(\frac{Y_{it}}{M_i} - \frac{\mu_{it}}{M_i} \right) = 0. \tag{E.1}$$

A two-period binary-treatment benchmark makes the estimand transparent. Suppose untreated expected patenting is $M_i a_t$ in both groups, and treatment multiplies firm i's post-period mean by e^{δ_i}. A saturated weighted PPML model matches group–period means of Y_{it}/M_i. Its treatment coefficient satisfies

$$e^{\beta} = \frac{\mathbb{E}[Y_{i1}/M_i \mid D_i = 1]/\mathbb{E}[Y_{i0}/M_i \mid D_i = 1]}{\mathbb{E}[Y_{i1}/M_i \mid D_i = 0]/\mathbb{E}[Y_{i0}/M_i \mid D_i = 0]} = \mathbb{E}[e^{\delta_i} \mid D_i = 1]. \tag{E.2}$$

Thus $\beta = \log \mathbb{E}[e^{\delta_i} \mid D_i = 1]$, the log of the average multiplicative response. By Jensen's inequality, $\beta \geq \mathbb{E}[\delta_i \mid D_i = 1]$, so when $\beta < 0$, $-\beta$ provides a lower bound on the magnitude of the average log decline. Unweighted PPML instead averages these responses with initial patent counts as a weight.

The empirical models additionally condition on fixed effects and scale responses by exposure. Within firms, inverse firm-size weights preserve technologies' relative sizes while adjusting each firm's overall contribution. Equation (E.2) gives the binary-treatment benchmark for this choice.

Figure E5 compares unweighted total-patenting responses with estimates that downweight only the largest fifth of portfolios, holding the sample, exposure scaling, and fixed effects constant. The low-share decline persists under both schemes. Among high-share firms, however, the unweighted comparison exhibits a pronounced pre-treatment divergence driven by Sindoh's matched set; both this divergence and the subsequent decline disappear when that set is omitted.[13] This pre-existing divergence prevents interpreting the unweighted high-share decline as an effect of liberalization. Downweighting only the largest fifth recovers the preferred specification's pattern of sustained patenting among high-share firms and declining patenting among low-share firms.

The within-firm estimates show similar reallocation patterns across weighting schemes, with a larger shift away from treated technologies among high-share firms (Figure E6). The unweighted estimates have larger standard errors, particularly in the high-share group.

[13] Omitting Sindoh's matched set reduces the absolute difference between the high-share coefficients for 1993 and 1997 from 0.398 to 0.023. The high-share mean of the 1998–2007 coefficients changes from −0.308 to 0.111, while the low-share mean remains almost unchanged, at −0.363 and −0.361.

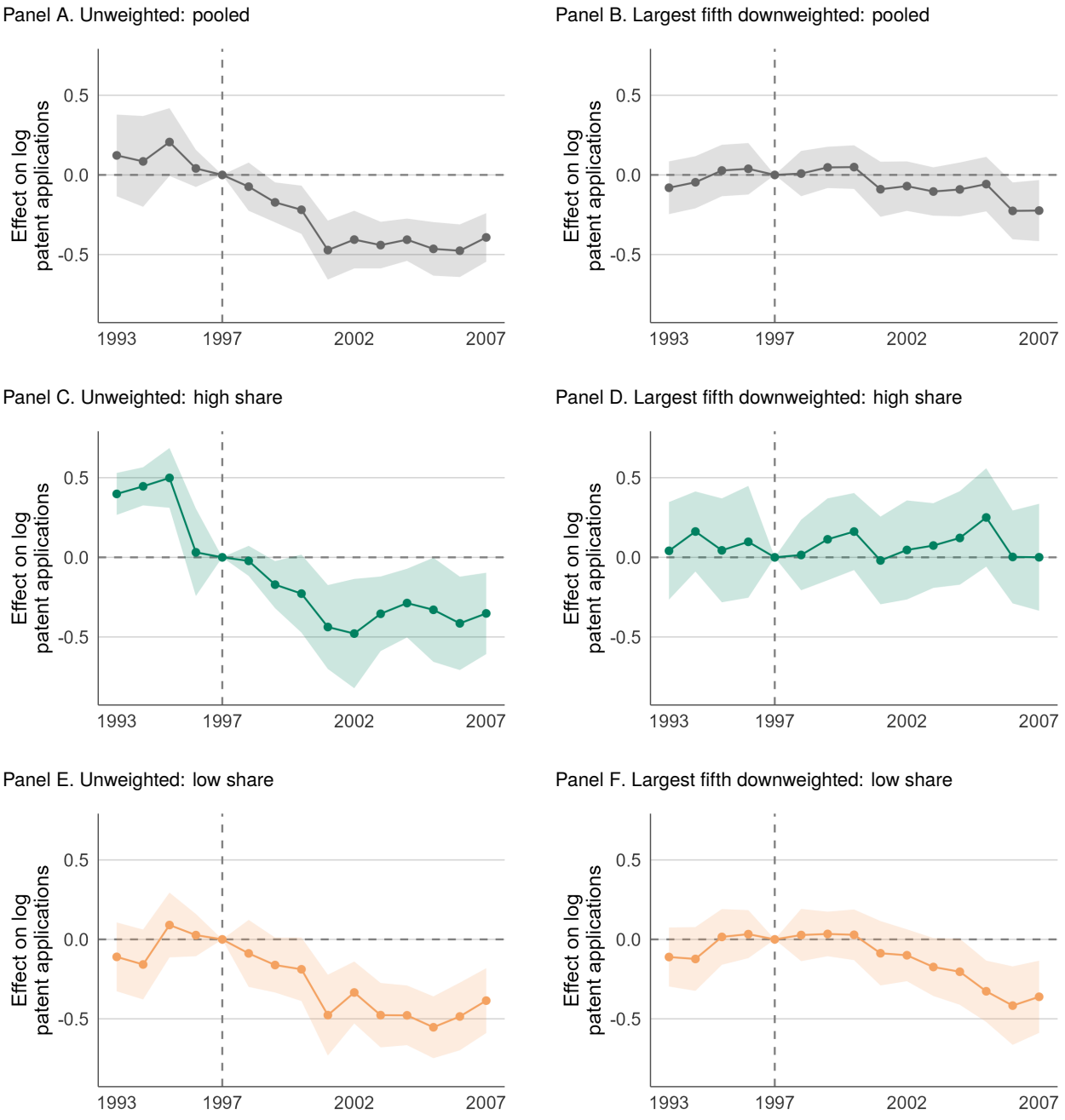


FIGURE E5. WEIGHTING AND TOTAL PATENTING

Notes: The panels compare total-patenting responses under two weighting schemes. The low-share decline persists, while downweighting the largest portfolios removes the pronounced high-share pre-treatment divergence and recovers the baseline pattern. Left panels omit patent-count weights; right panels assign $\min\{1, q_{0.8}/M_i\}$, where M_i is the 1985–1997 patent count and $q_{0.8}$ its 80th percentile among distinct matched firms. Both retain the matching-reuse adjustment. Rows show pooled, high-share, and low-share responses. Pooled treatment-by-post coefficients (firm-clustered standard errors) are -0.417 (0.139) and -0.058 (0.079). Both specifications use 11,323 observations and 697 firm clusters (238 exposed firms and 459 controls). Exposure scaling, fixed effects, and the share cutoff follow Figure 6(A). Bands are 90 percent confidence intervals relative to 1997.

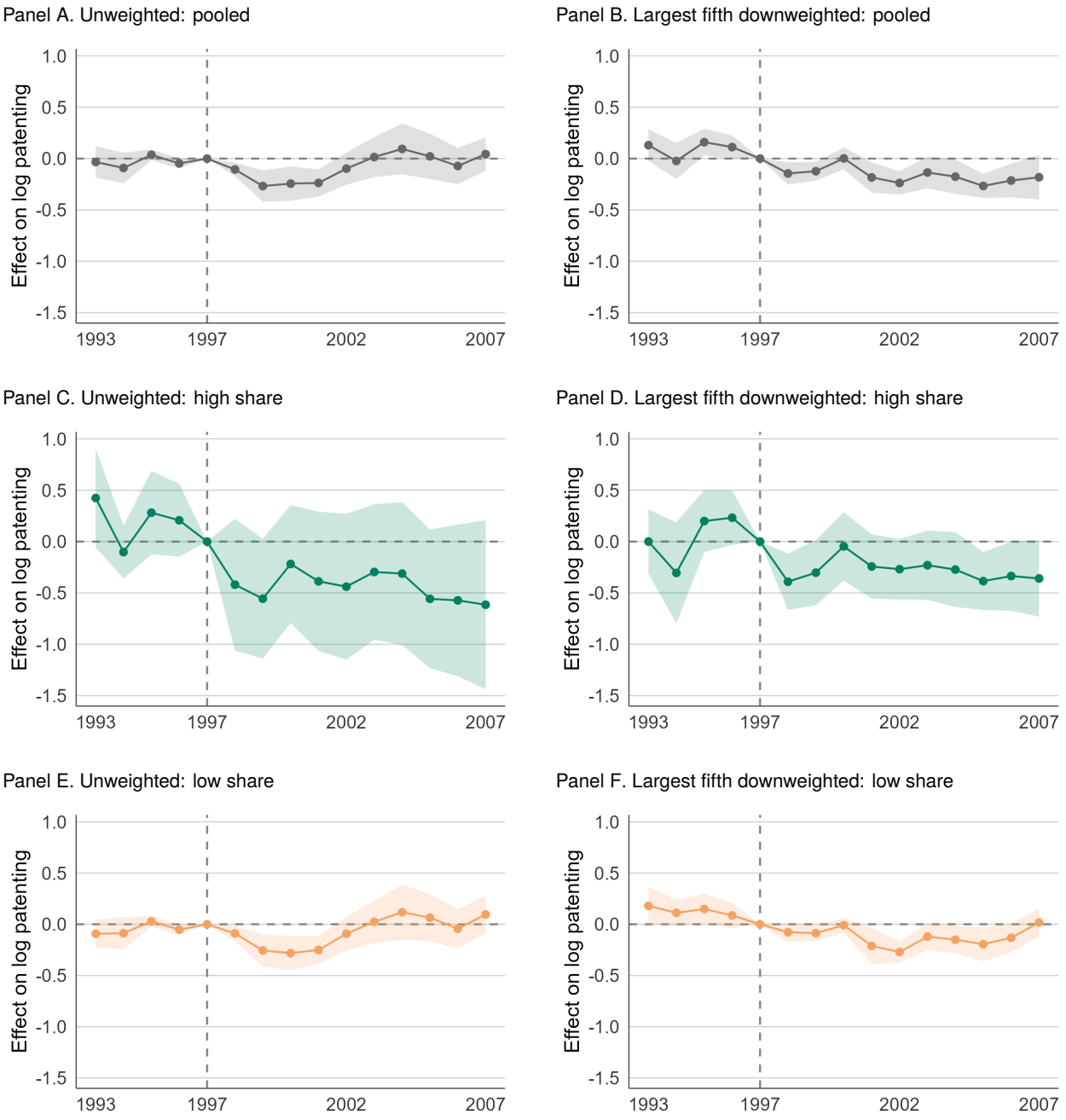


FIGURE E6. WEIGHTING AND WITHIN-FIRM REALLOCATION

Notes: The panels compare treated and untreated patenting within firms under two weighting schemes. High-share firms exhibit a larger estimated retreat from treated technologies under both schemes; downweighting the largest portfolios narrows their confidence intervals. Left panels assign equal firm weights; right panels assign $\min\{1, q_{0.8}/M_i\}$, where M_i is the 1985–1997 patent count and $q_{0.8}$ its 80th percentile among sample firms. Rows show pooled, high-share, and low-share responses. Pooled treatment-by-post coefficients (firm-clustered standard errors) are -0.050 (0.050) and -0.198 (0.054). Both specifications use 100,955 observations and 504 firm clusters. Exposure scaling, fixed effects, and the share cutoff follow Figure 6(B). Bands are 90 percent confidence intervals relative to 1997.

E.4 Exposure and Patent Activity

Table E1 examines exposure scaling and continued patenting.

Binary treatment interactions examine average responses without scaling by exposure. The within-firm coefficients remain negative, with a greater retreat among high-share firms. This check shows that reallocation also appears without exposure scaling.

To assess whether the results depend on firms ceasing to patent, I retain applicants with a filing in 2007 or later and repeat matching. Both share groups retreat from treated technologies, while low-share firms also reduce total patenting (Figure E7). The pattern thus persists among continuing applicants.

TABLE E1—ROBUSTNESS OF TOTAL PATENTING AND WITHIN-FIRM REALLOCATION

	Between firms			Within firms		
	Pooled	High share	Low share	Pooled	High share	Low share
Binary treatment	-0.012	0.201	-0.276*	-0.747***	-1.214***	-0.386***
	(0.124)	(0.171)	(0.166)	(0.126)	(0.212)	(0.139)
Firms / clusters	697	697	697	504	504	504
Observations	11,323	11,323	11,323	100,955	100,955	100,955
Continuing applicants	-0.108*	0.063	-0.253***	-0.368***	-0.503***	-0.221**
	(0.057)	(0.086)	(0.071)	(0.083)	(0.123)	(0.090)
Firms / clusters	353	353	353	300	300	300
Observations	6,841	6,841	6,841	91,897	91,897	91,897

Notes: Each entry is the coefficient on a single treatment-by-post interaction in the corresponding PPML specification for 1993–2007; the post period is 1998–2007. Within-firm estimates show a relative retreat from treated technologies across the specifications. Pooled responses are estimated separately; high- and low-share responses are estimated jointly. Columns use equations (5) and (6), respectively, replacing annual treatment interactions with post interactions. All specifications use inverse pre-period patent-count weights; between-firm weights additionally adjust for reuse across matched sets. Binary treatment replaces exposure-scaled interactions. Continuing applicants have a filing in 2007 or later; the between-firm sample is rematched. The share cutoff and exposure reference remain 70.79 and 10.75 percent. Parentheses report firm-clustered standard errors. Firm counts refer to each full regression, including both share groups in joint models. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

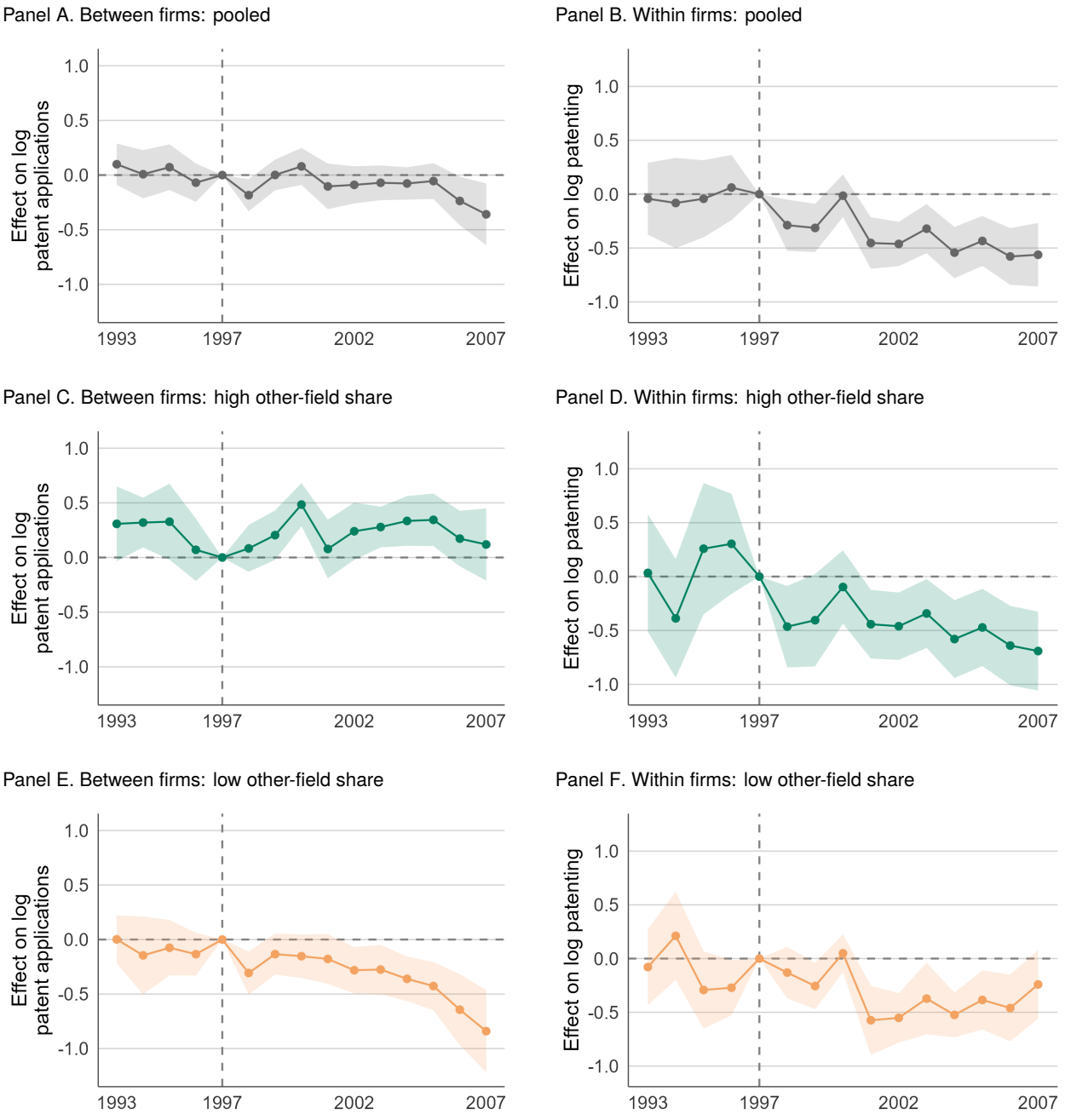


FIGURE E7. FIRM RESPONSES AMONG CONTINUING PATENT APPLICANTS

Notes: The panels estimate responses among firms with a patent or utility-model filing in 2007 or later, using records for 1985–2015. Low-share firms reduce total patenting, and both groups shift away from treated technologies. The between-firm analysis applies this requirement to exposed firms and controls before repeating the main matching procedure; the estimation sample contains 131 exposed firms (66 high-share and 65 low-share) and 222 controls in 131 matched sets. The within-firm analysis restricts the main sample to 300 firms (154 high-share and 146 low-share). Weights based on inverse pre-period patent counts, exposure scaling, fixed effects, the share cutoff, and the reference exposure follow the main specifications. Rows report pooled, high-share, and low-share responses. Bands are 90 percent firm-clustered confidence intervals, relative to 1997.

E.5 Initial Firm Size and the Other-Field-Share Split

The low-share group includes more large firms. In the within-firm sample, the median low-share firm filed 15 patents in 1985–1997, compared with 10 for high-share firms, and the ratio of the 80th to the 20th percentile is 33.8 against 7.1 (Table E2). Revenue, employment, leverage, liquidity, and ROA are observed only for the 211 sample firms matched to 1997 ValueSearch financial statements, not for the full estimation sample.[14] Among these firms, median 1997 revenue is 187 billion won in the low-share group and 37 billion won in the high-share group. Excluding firms with more than 100, 50, or 20 pre-period patents brings the two groups closer on every size measure, and under the cap of 20 their patent counts are broadly similar. Revenue and employment remain somewhat larger in the low-share group under each cap; under the cap of 20, median revenue is 30 and 24 billion won and median employment is 237 and 212. Leverage is slightly higher in the low-share group in every sample, while liquidity is similar across groups.

Table E2—Firm Size and Financial Characteristics by Other-Field Share

	Full sample		≤100		≤50		≤20	
	High	Low	High	Low	High	Low	High	Low
Patents	10	15	9	9	9	8	7	7
	(7.1)	(33.8)	(5.2)	(7.3)	(4.5)	(7.0)	(4.3)	(4.0)
IPC subclasses	7	8	7	6	6	5	6	5
	(3.9)	(12.2)	(3.4)	(4.3)	(3.0)	(3.3)	(2.5)	(2.5)
Revenue	37	187	30	65	27	65	24	30
	(15.2)	(45.7)	(11.6)	(13.5)	(10.7)	(12.7)	(9.7)	(18.1)
Employees	273	909	263	333	235	324	212	237
	(10.5)	(28.5)	(9.3)	(8.9)	(9.0)	(9.0)	(6.2)	(8.6)
Leverage	0.71	0.77	0.70	0.74	0.70	0.73	0.72	0.78
	(1.5)	(1.4)	(1.5)	(1.6)	(1.5)	(1.6)	(1.5)	(1.7)
Liquidity	1.10	1.01	1.11	1.13	1.11	1.13	1.11	1.11
	(2.0)	(1.9)	(2.0)	(2.2)	(2.0)	(2.1)	(2.1)	(2.0)
ROA (%)	7.1	7.2	7.2	7.1	7.2	7.1	6.5	6.9
	(3.2)	(3.0)	(3.3)	(3.4)	(3.2)	(3.3)	(3.2)	(4.5)
Firms	253	251	238	195	225	175	187	139
with financials	105	106	94	71	88	61	63	43

Notes: The sample is the within-firm sample of Figure 7, split at the 70.79 percent other-field share. Columns keep firms whose 1985–1997 patent count does not exceed the stated cap. Entries are group medians; parentheses below report the ratio of the 80th to the 20th percentile. Patents are 1985–1997 patent counts; IPC subclasses count four-character IPC subclasses with positive 1985–1997 patent counts. Revenue, employees, leverage, liquidity, and ROA are 1997 values from ValueSearch financial statements and cover only the sample firms matched to ValueSearch with 1997 data, mostly externally audited firms (last row). Revenue is in billions of won. Leverage is total liabilities divided by total assets; liquidity is the current ratio, current assets divided by current liabilities; ROA is operating income divided by total assets.

[14] Match rates are similar across the two groups: 105 of 253 high-share and 106 of 251 low-share firms in the full sample, and 63 of 187 and 43 of 139 under the cap of 20. Differences in coverage are therefore unlikely to drive the financial comparisons.

Table E3 repeats the portfolio-split estimates under the same restrictions. In each restricted sample, high-share firms sustain total patenting and retreat more strongly from treated technologies, and the high-minus-low differences remain close to their full-sample values. The heterogeneity in Figure 7 therefore does not appear to arise from the concentration of large firms in the low-share group.

Table E3—Portfolio-Split Responses in Samples Restricted by Initial Patenting

	Full sample	≤100	≤50	≤20
Total applications, between firms				
High share	0.218	0.244*	0.249*	0.271**
	(0.133)	(0.132)	(0.133)	(0.133)
Low share	-0.190*	-0.129	-0.140	-0.151
	(0.100)	(0.104)	(0.116)	(0.130)
High − low	0.408**	0.372**	0.389**	0.422**
	(0.160)	(0.161)	(0.170)	(0.180)
Exposed firms (H / L)	120 / 118	118 / 103	115 / 93	103 / 70
Treated technologies, within firms				
High share	-0.503***	-0.537***	-0.543***	-0.618***
	(0.155)	(0.159)	(0.159)	(0.171)
Low share	-0.164*	-0.160	-0.173	-0.151
	(0.092)	(0.101)	(0.125)	(0.140)
High − low	-0.339*	-0.377**	-0.371*	-0.467**
	(0.180)	(0.189)	(0.204)	(0.226)
Firms (H / L)	253 / 251	238 / 194	225 / 174	185 / 138

Notes: Entries are mean annual 1998–2007 event-study coefficients from the specifications of Figure 7; the high − low row is their difference, with standard errors from the full covariance matrix. Columns restrict 1985–1997 patent counts to the stated cap. The between-firm columns drop matched sets whose exposed firm exceeds the cap; matched controls, weights, and fixed effects follow the main specification. The within-firm columns drop firms above the cap. The 70.79 percent share cutoff and the 10.75 percent exposure reference are held fixed. Parentheses report firm-clustered standard errors. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E.6 Influence of Individual Firms and Products

A single large firm or liberalized product could drive apparent reallocation. Figures E8 and E9 assess this possibility by removing one matched set or firm, and one treated product, at a time while fixing the share cutoff and exposure reference. The within-firm response remains negative with a 90 percent interval below zero in both share groups under every omission. The stronger retreat among firms with greater outside-field experience also persists. This pattern is shared across firms and components of the policy shock.

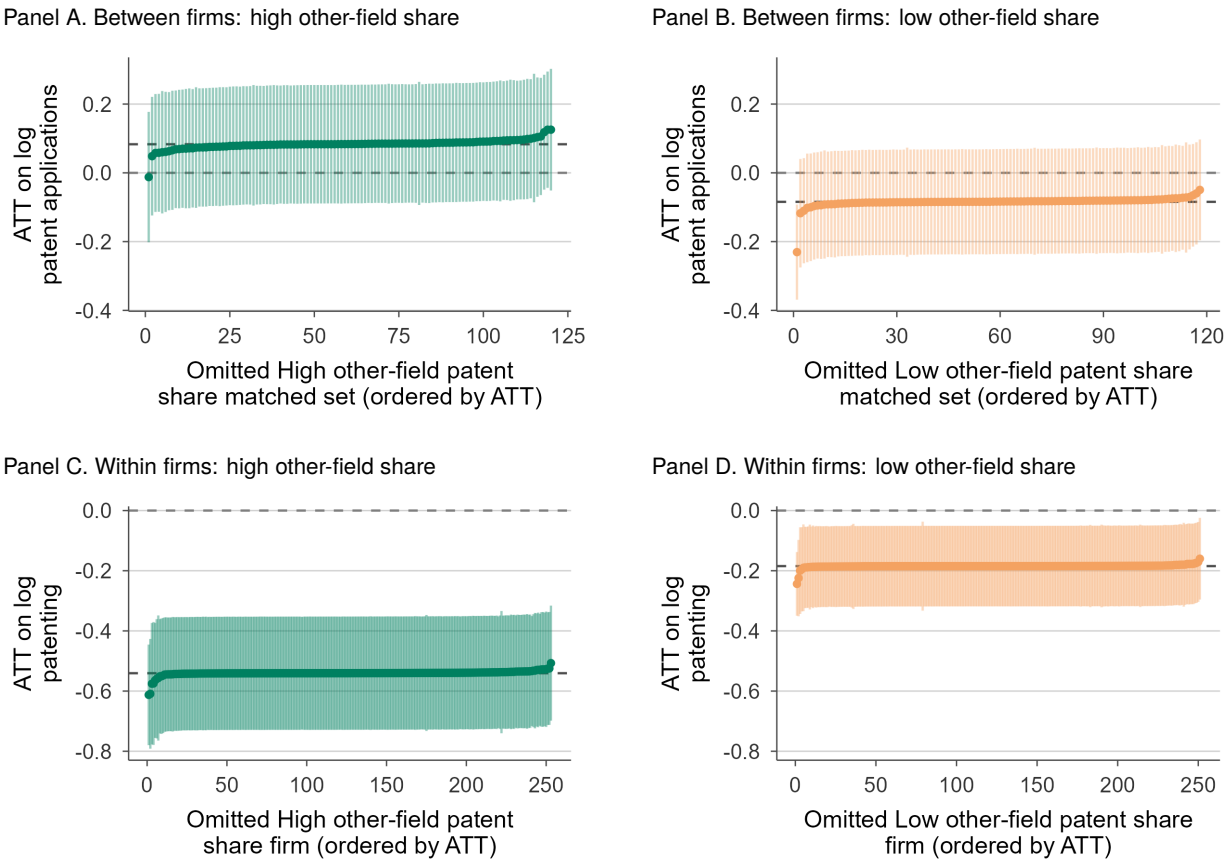


FIGURE E8. INFLUENCE OF INDIVIDUAL MATCHED SETS AND FIRMS

Notes: Each point is the single treatment-by-post coefficient after one omission, ordered by the resulting estimate. Vertical bars are 90 percent firm-clustered confidence intervals; the horizontal reference marks the corresponding full-sample coefficient. The split cutoff and exposure reference are fixed. Matched-set omissions remove the exposed firm and its matched set; within-firm omissions remove one firm. The high- and low-share within-firm coefficients range from -0.613 to -0.507 and from -0.244 to -0.160, respectively.

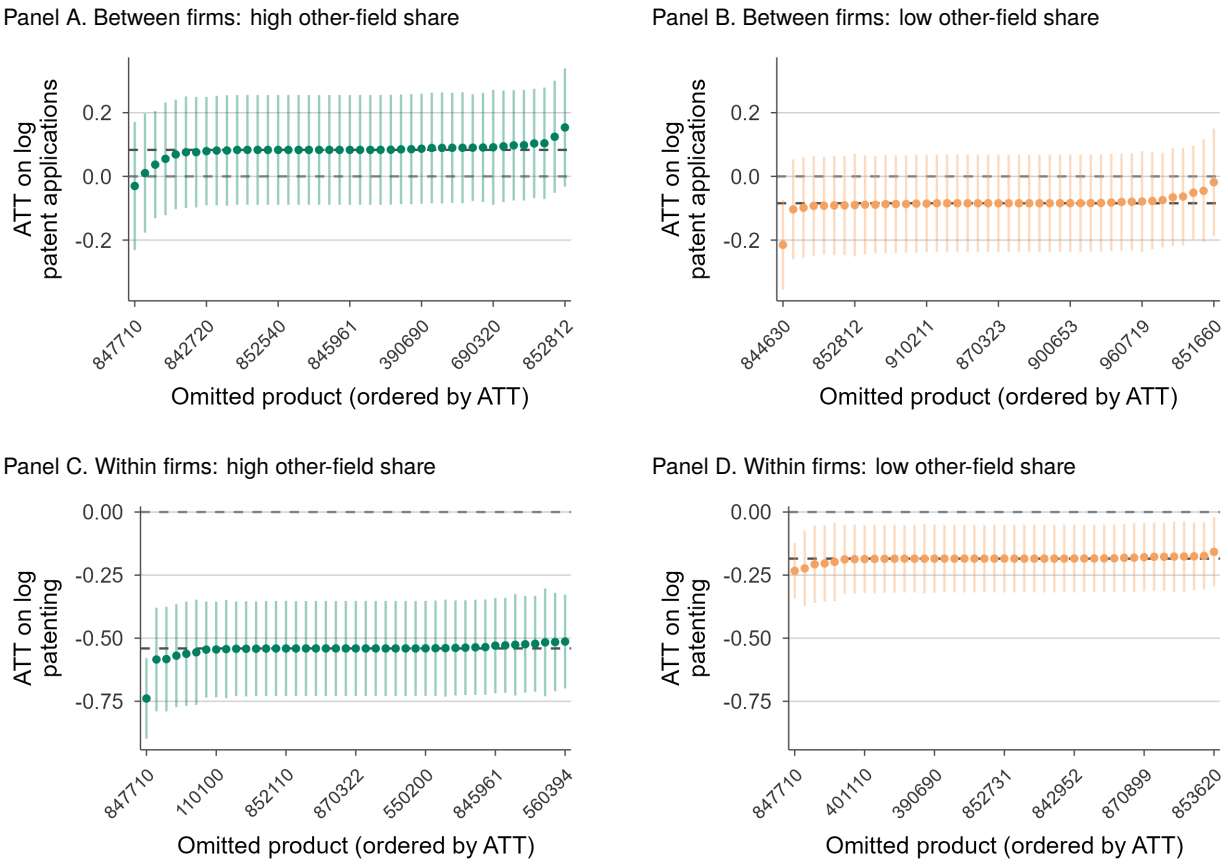


FIGURE E9. INFLUENCE OF INDIVIDUAL TREATED PRODUCTS

Notes: Each point is the single treatment-by-post coefficient after one omission, ordered by the resulting estimate. Vertical bars are 90 percent firm-clustered confidence intervals; the horizontal reference marks the corresponding full-sample coefficient. The split cutoff and exposure reference are fixed. Product omissions reclassify cells linked only to the omitted product and recompute exposure. The between-firm analysis reselects and rematches firms; the within-firm analysis retains eligible firms from the main sample.

E.7 Destination Characteristics and Portfolio Shares

For firm i and IPC four-character subclass k, let s_{ik}^{pre} and s_{ik}^{post} denote its shares of total patent counts in 1988–1997 and 1998–2007. Table 4 estimates, separately by exposure group,

$$100(s_{ik}^{\text{post}} - s_{ik}^{\text{pre}}) = \alpha_i + Z_{ik}'\gamma + b_1(100s_{ik}^{\text{pre}}) + b_2(100s_{ik}^{\text{pre}})^2 + b_3\mathbf{1}\{s_{ik}^{\text{pre}} = 0\} + \varepsilon_{ik}. \tag{E.3}$$

Columns (1)–(5) enter one characteristic at a time; column (6) enters all five. Regressions are unweighted, with standard errors clustered by firm and IPC subclass. Exposed firms have exposure of at least 5 percent; low-exposure firms have exposure at most 1 percent. Both have a pre-period patent count of at least five and positive total patenting in each ten-year window. Table 4 reports the resulting samples.

Own relatedness measures a destination's similarity to the firm's 1985–1997 portfolio outside that destination. Similarity is based on cosine proximity in the domestic patent co-classification network and is averaged using the firm's portfolio weights. Domestic growth is the log ratio of Korean corporate patent counts in 1993–1997 to that in 1988–1992. Foreign growth is the analogous log ratio of publication-by-CPC-symbol counts pooled across the US, European, Japanese, and Chinese patent offices. Domestic and foreign stocks are the logs of cumulative publication-by-symbol counts in 1988–1997 at the Korean office and the four foreign offices, respectively. Office counts include applicants of all nationalities and can count several symbols per document.

To make coefficients comparable across characteristics and exposure groups, I standardize each characteristic using a common mean and standard deviation across eligible firm–IPC pairs. The regressions use the subset with all required characteristics observed, and coefficients measure percentage-point changes per common standard deviation.

E.8 Destination Characteristics in the Within-Firm Event Study

The complementary event study extends Figure 8. For a destination characteristic Z_{ic}, I estimate

$$\log \mu_{ict} = \alpha_{ic} + \lambda_{it} + \eta_{g(c)t} + \sum_{s \neq 1997} \mathbf{1}\{t = s\} \left[\frac{E_i}{\bar{E}} \{a_s S_{ic} + \beta_s O_{ic} + \gamma_s O_{ic} Z_{ic}\} + \pi_s Z_{ic} \right], \tag{E.4}$$

where S_{ic} and O_{ic} indicate untreated same-field and other-field cells. The Z-by-year terms allow technologies with different pre-period characteristics to follow separate paths independently of firm exposure. The

time-invariant level of Z is absorbed by firm–cell effects. The coefficient β_s measures the response in other fields relative to treated technologies at the mean of Z; γ_s measures how this response varies with destination characteristics.

The event studies use the 504 firms, reference exposure, weights, and fixed effects of the main within-firm analysis. Each characteristic is assigned at the IPC-subclass level and standardized across other-field firm–cell pairs. The same centering and scaling are applied to same-field and treated cells. Figure E10 shows the annual coefficients for foreign growth. The positive post-period gradient is consistent with adjustment toward technologies growing abroad.

Figure E11 estimates the model separately for domestic growth, foreign growth, domestic stock, and foreign stock, and averages each γ_s over 1998–2007. Foreign growth has the largest positive point estimate. The intervals use the full covariance matrix of the annual coefficients.

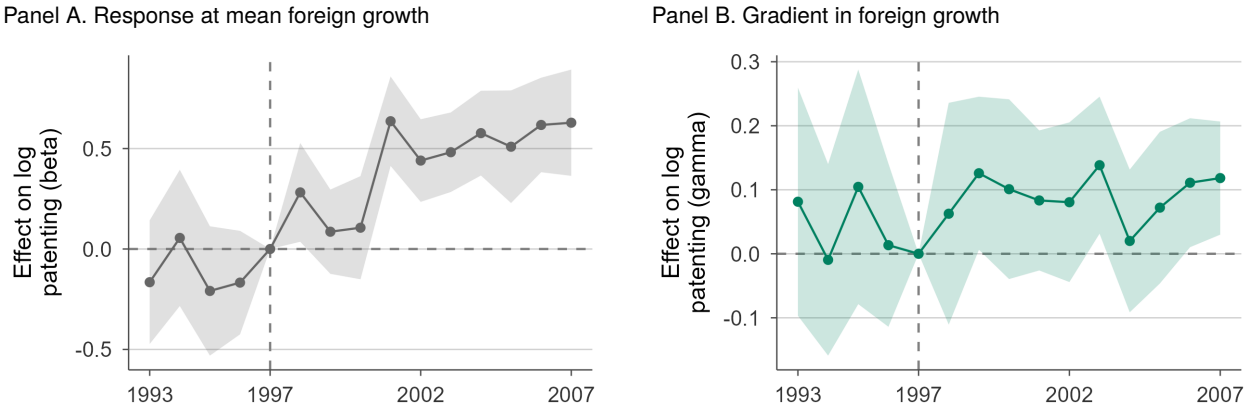


FIGURE E10. FOREIGN TECHNOLOGICAL GROWTH AND WITHIN-FIRM REALLOCATION

Notes: Annual coefficients β_s and γ_s from equation (E.4), relative to 1997. The estimated post-period gradient in foreign growth is positive. Mean annual post-period coefficients (clustered standard errors in parentheses) are 0.436 (0.112) for β_s and 0.091 (0.056) for γ_s; the latter has $p = 0.103$. Growth compares 1988–1992 and 1993–1997. The sample contains 504 firms; weights based on inverse pre-period patent counts and the 10.75 percent exposure reference follow the main specification. Bands are 90 percent firm-clustered confidence intervals.

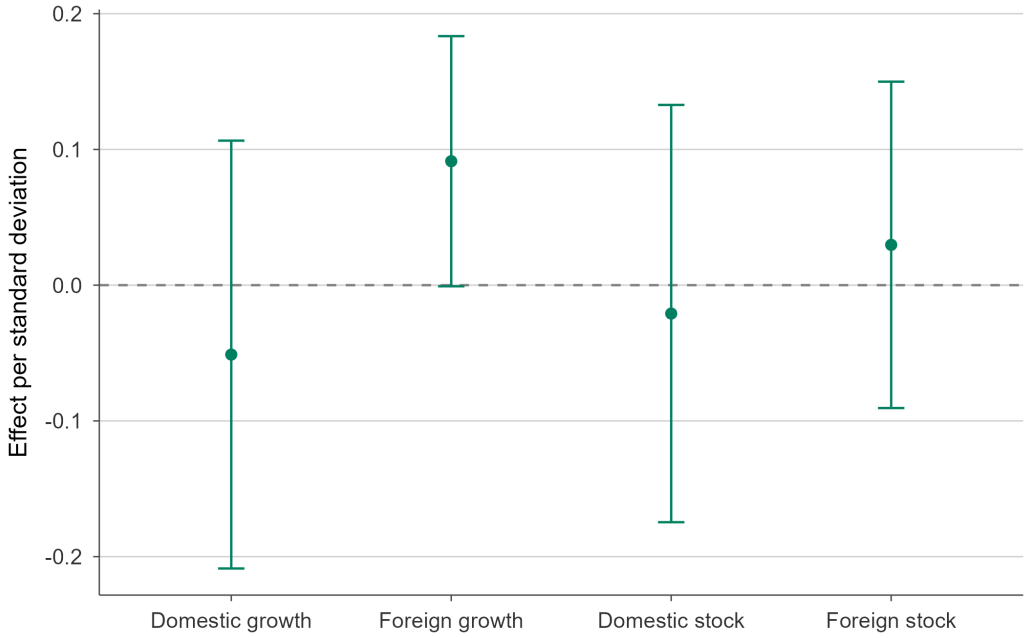


FIGURE E11. POST-LIBERALIZATION GRADIENTS IN DESTINATION CHARACTERISTICS

Notes: Each point averages the annual destination gradient in 1998–2007 from a separate version of equation (E.4). Growth compares 1988–1992 and 1993–1997; stocks cover 1988–1997. Characteristics are standardized across other-field firm–cell pairs. Vertical intervals are 90 percent firm-clustered confidence intervals calculated from the full covariance matrix.

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